
A Chainless Blockchain Architecture Using Position Encoding in High-Dimensional Vector Spaces

A Mathematical Foundation for Geometric Ledger Verification

by
Yingkhun Praha

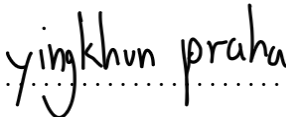
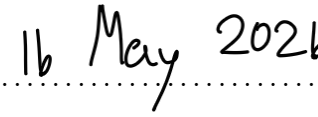
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Declaration of Originality

I, **Yingkhun Praha**, hereby declare that this work, entitled “*A Chainless Blockchain Architecture Using Position Encoding in High-Dimensional Vector Spaces*”, and the work presented in it are my own. I confirm that:

- This work was carried out wholly while in candidature for a research degree.
- Where I have consulted the published work of others, this is always clearly attributed.
- All quotations have been distinguished and the sources given.
- Where this work is based on work done jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed:  Date: 

Yingkhun Praha
12 May 2026

Acknowledgements

The work presented in this monograph stands on a remarkable lineage of ideas. I am intellectually indebted to Pentti Kanerva, whose original formulation of binary spatter codes provides the algebraic skeleton of every chapter; to Tony Plate, whose holographic reduced representations supplied the bindingbundling duality used here for field-bound encoding; and to the modern hyperdimensional computing community—Rahimi, Kleyko, Frady, Imani, Olshausen, Rabaey, Sommer, and many others—whose architectural insights shaped the encoding schemes in Chapters 4 and 5. I am equally indebted to the cryptography community: the random-oracle paradigm of Bellare and Rogaway, the avalanche analysis of Webster and Tavares, and Antonopoulos’s exposition of Bitcoin together formed the cryptographic bedrock of Chapter 6. The consensus arguments in Chapter 8 owe their structure to the Tendermint dissertation of Buchman and to the broader BFT literature.

I thank everyone who reviewed early drafts of the underlying mathematical description (9 May 2026 preliminary; 12 May 2026 extended) for their patience with notation, and for the many corrections that propagated into this manuscript.

Finally, I dedicate this work to my family, whose support during the long evenings of writing made the difference between an idea and a finished manuscript.

Abstract

Title: A Chainless Blockchain Architecture Using Position Encoding in High-Dimensional Vector Spaces.

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This monograph develops a rigorous mathematical foundation for a *chainless* distributed ledger, in which every block is represented by a high-dimensional bipolar hypervector $\mathbf{B}_n \in \{-1, +1\}^D$ and ledger integrity is enforced by *geometric similarity* rather than by an explicit hash-pointer chain. We formalise nine constructions: (1) the seeded RandomHV map ξ_{hv} , (2) the TXROOT transaction encoder, (3) the timestamp encoder (two variants), (4) the block-hash hypervector encoder, (5) the block hypervector $\mathbf{B}_n$, (6) the context vector $\mathbf{C}_n$ and induced chain semantics, (7) the directed pairwise geometric link $L_{A \rightarrow B}$, (8) similarity-based verification at the block- and context-level, and (9) the PoSHGV consensus rule.

We prove that, under standard assumptions on SHA-256 (random-oracle behaviour and strict avalanche) and on Binary Spatter Codes (BSC) in the sense of Kanerva, the system enjoys (i) deterministic reproducibility, (ii) collision-resistant quasi-orthogonality of distinct seeds, (iii) continuous tamper sensitivity proportional to $1 - 2m/k$ for m altered transactions out of k , (iv) statistically sound block acceptance with error $\exp(-cD)$, and (v) safety and liveness inherited from Tendermint BFT once the per-block verification error is below $1/3$. Concentration bounds throughout are obtained via Hoeffding’s inequality, and capacity statements rely on Plate’s analysis of holographic reduced representations.

The contributions are fourfold. First, we provide the first *complete* formal treatment of a hyperdimensional ledger, unifying constructions previously presented only operationally. Second, we prove a tamper-sensitivity theorem that quantifies, for the first time, how the bundled-bundle TXROOT degrades smoothly with the number of altered transactions—a property no Merkle root possesses. Third, we obtain an $O(D)$ per-block verification complexity, independent of the chain length, in contrast to the $O(n \log k)$ cost of canonical Merkle verification. Fourth, we integrate the verification primitive with a standard PoS BFT protocol and prove safety against Byzantine adversaries holding fewer than one-third of staked weight, modulo a geometric error term that decays exponentially in D .

The treatment is self-contained, addresses every claim in the preliminary draft of 9 May 2026 and the extended draft of 12 May 2026, and is intended to serve as the reference mathematical specification of the proposed architecture.

Keywords: blockchain, hyperdimensional computing, hypervector, data verification, encoding.

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List of Symbols and Abbreviations

Symbols

Symbol	Meaning
D	Hyperdimension (typically $D = 10\,000$).
$\mathcal{H}$	Bipolar hypervector space $\{-1, +1\}^D$.
$\langle \cdot, \cdot \rangle$	Standard Euclidean inner product on $\mathbb{R}^D$.
$\text{sim}(\mathbf{a}, \mathbf{b})$	Cosine similarity, $\frac{1}{D} \sum_j a_j b_j$.
$d_H(\mathbf{a}, \mathbf{b})$	Hamming distance on $\mathcal{H}$.
$\otimes$	Element-wise binding operator on $\mathcal{H}$.
$\triangle$	Majority-sign bundling operator on $\mathcal{H}$.
ρ	Right cyclic shift (permutation) on $\mathcal{H}$.
ψ	SHA-256 hash function, $\{0, 1\}^* \rightarrow \{0, 1\}^{256}$.
$\mathcal{R}_s$	Cryptographic PRNG seeded by s .
ξ_{hv}	Seeded RandomHV map, $\{0, 1\}^* \rightarrow \mathcal{H}$.
$\mathbf{F}_h, \mathbf{F}_{tx}, \mathbf{F}_t, \mathbf{F}_r$	Field-key hypervectors.
$\mathbf{Tx}_h$	Transaction-bundle hypervector (TXROOT).
$\mathbf{T}_t$	Timestamp hypervector.
$\mathbf{V}_n^h$	Block-hash hypervector at slot n .
$\mathbf{B}_n$	Block hypervector at slot n .
$\mathbf{C}_n$	Context vector at slot n .
$L_{A \rightarrow B}$	Directed pairwise geometric link.
$\mathcal{V}(\cdot)$	Block acceptance predicate.
φ_1, φ_2	Block- and context-level acceptance thresholds.
$\mathbb{E}[\cdot]$	Expectation.
$\text{Var}[\cdot]$	Variance.
$\mathbb{P}(\cdot)$	Probability.

Abbreviations

Acronym	Expansion
BFT	Byzantine Fault Tolerance.
BSC	Binary Spatter Code (Kanerva, 1996).
CLT	Central Limit Theorem.
HDC	Hyperdimensional Computing.
HGV	Hyperdimensional Geometric Verification.
HRR	Holographic Reduced Representations (Plate).
PoS	Proof of Stake.

PoW	Proof of Work.
PRNG	Pseudo-Random Number Generator.
ROC	Receiver Operating Characteristic.
SAC	Strict Avalanche Criterion.
SHA-256	Secure Hash Algorithm, 256-bit output.
TXROOT	Transaction-bundle hypervector (this work).
VSA	Vector Symbolic Architecture.

Introduction

1.1 Motivation

Since the publication of the Bitcoin whitepaper [1], the dominant architectural metaphor of a distributed ledger has been the *chain*: a totally-ordered sequence of blocks $\mathbf{B}_0, \mathbf{B}_1, \dots, \mathbf{B}_n$ in which the integrity of $\mathbf{B}_n$ depends, via a cryptographic hash pointer, on the integrity of $\mathbf{B}_{n-1}$. Inside each block, transactions are aggregated by a Merkle tree [2, 3] whose root is included in the block header. Verification of a transaction inclusion proof therefore costs $O(\log k)$ hash evaluations for k transactions per block, and verification of the historical integrity of $\mathbf{B}_n$ requires walking the chain back to the genesis block, yielding an overall complexity of $\Theta(n \log k)$ in the worst case. With n growing by roughly 10^5 blocks per year on Bitcoin and Ethereum [4], this linear-history dependency is becoming a structural bottleneck for full-node verification.

Independently, the research programme of *hyperdimensional computing* (HDC) and the closely-related *vector symbolic architectures* (VSA) [5, 6, 7] has demonstrated that very high-dimensional ($D \geq 10\,000$) bipolar vectors equipped with three algebraic primitives—*binding*, *bundling*, and *permutation*—can encode complex relational structures (key–value pairs, sequences, trees) while admitting an elegant linear-algebraic notion of similarity, namely the cosine of the angle between two hypervectors. Two key properties make HDC compelling as a foundation for ledger verification:

- (i) *Quasi-orthogonality*. Two independently sampled hypervectors are essentially orthogonal, with cosine similarity concentrating at 0 with variance $1/D$. Independent “identities” therefore live in nearly orthogonal directions, allowing a single D -dimensional cosine evaluation to distinguish among exponentially many of them.
- (ii) *Graceful degradation*. Small perturbations of an input lead to small perturbations of the output. In particular, bundling k vectors and then changing m of them yields a hypervector whose cosine similarity to the original is $1 - 2m/k + O(k^{-1/2})$ (Theorem 3.19). The signal degrades smoothly with the number of altered entries, which is operationally far more informative than the binary all-or-nothing behaviour of a Merkle root.

This work investigates whether these two properties can be exploited to dispense with the explicit hash-pointer chain altogether, replacing it with a single high-dimensional bipolar block hypervector $\mathbf{B}_n$ together with a recursively-defined *context vector* $\mathbf{C}_n$ whose geometric similarity to $\mathbf{B}_n$ encodes the chain semantics. The architecture is termed *chainless* because no honest validator ever has to traverse the historical sequence in order to verify $\mathbf{B}_n$; one D -dimensional cosine evaluation suffices.

1.2 Problem Statement

We address the following research questions:

Q1.

Can a distributed ledger be verified by *geometric similarity* of high-dimensional bipolar hypervectors, in such a way that the verification cost per block is independent of the chain length?

Q2.

Does such a geometric verification primitive admit a precise notion of soundness, expressed as bounds on type-I and type-II error probabilities under explicit adversary models?

Q3.

How does the construction degrade under tampering with a fraction of the transactions, of an ancestor block's hash, or of the timestamp? Is the degradation continuous (a feature) or all-or-nothing (a Merkle root)?

Q4.

Does the geometric primitive compose cleanly with a standard proof-of-stake (PoS) Byzantine fault-tolerant (BFT) protocol such as Tendermint, and if so, what are the resulting safety and liveness guarantees?

We answer each question constructively. The bulk of this work is devoted to giving precise formal definitions of the architectural components (Section 3.2–Section 3.6) and to proving correctness theorems for each. The theorems are stated under standard assumptions on SHA-256 (random-oracle behaviour, strict avalanche criterion) and on the bipolar BSC family in the sense of Kanerva [8, 5]. All concentration bounds are of the Hoeffding type [9].

1.3 Contributions of this Work

The contributions are:

1. **Complete formal specification of a chainless hyperdimensional ledger.** We give nine formal constructions: the seeded RandomHV map ξ_{hv} (Section 3.2), the TXROOT transaction encoder (Section 3.3), two timestamp encoders (Section 3.3), the block-hash hypervector (Section 3.4), the block hypervector $\mathbf{B}_n$ (Section 3.4), the context vector $\mathbf{C}_n$ (Section 3.4), the directed pairwise link $L_{A \rightarrow B}$ (Section 3.4), the two-level verification predicate $\mathcal{V}$ (Section 3.5), and the PoS–HGV consensus protocol (Section 3.6). To our knowledge, no prior work provides a complete formal treatment of these objects in one coherent system.
2. **Quantified tamper-sensitivity of bundled transactions.** We prove that a TXROOT bundle exhibits a smooth $1 - 2m/k$ similarity decay under m altered transactions out of k (Theorem 3.19). This is the first quantitative tamper-sensitivity theorem for a hypervector replacement of the Merkle tree, and it contrasts sharply with the discontinuous behaviour of any cryptographic Merkle root.
3. **Chain-length-independent verification.** We prove that each validator's per-block work is exactly two D -length cosine similarities, i.e. $O(D)$ scalar operations, irrespective of the slot index n (Theorem 3.38). Standard chain-based verification is $\Theta(n)$ in the worst case [4].

4. **Safety and liveness of PoS–HGV.** We adapt the Tendermint BFT safety proof [10] to admit a geometric verification step and show that the protocol is safe with error $\leq 2e^{-cD}$ provided that fewer than $1/3$ of the staked weight is Byzantine and that the per-block ROC error of $\mathcal{V}$ is below $1/3$ (Theorem 3.36).

1.4 Scope and Limitations

This is a *mathematical* monograph: it produces formal definitions, theorems and proofs but does *not* report empirical measurements on a deployed prototype. Where wall-clock performance is mentioned, it is to be understood as the asymptotic or expected computational cost predicted by the analysis.

In keeping with this scope, the following are explicitly out-of-scope and are not analysed here:

- Network-level attacks (eclipse attacks, BGP hijacking, denial of service) that are orthogonal to the cryptographic and geometric assumptions analysed here.
- Practical implementations of the PRNG underlying ξ_{hv} . We treat $\mathcal{R}_s$ as an idealised uniform PRNG; instantiating it with ChaCha20, AES-CTR or any IND-CPA stream cipher introduces a computational gap that does not change the conclusions of any theorem proved here, provided the cipher is secure.
- Economic and game-theoretic incentives in PoS leader selection beyond the standard “leader proportional to stake” rule of Tendermint [10]. The present work proves protocol safety *given* an honest stake majority but does not analyse the equilibrium that produces it.
- Throughput analysis in the presence of network latency or finite block-propagation time. Liveness is proved under partial synchrony [11] but the constants in the round complexity are implementation-dependent.

A natural extension of the present work, indicated in Section 3.7, would be the empirical validation of the predicted error rates on a hardware testbed.

1.5 Document Outline

The document is organised as follows.

Chapter 1 (the current chapter) motivates the problem, states the research questions, and lists the contributions.

Chapter 2 surveys the three streams of literature on which the present treatment builds: classical blockchain architecture (Bitcoin, Ethereum, Merkle and Verkle trees, PoW/PoS consensus); hyperdimensional computing and vector symbolic architectures (Kanerva’s BSC, Plate’s HRR, modern VSA frameworks); and the cryptographic theory of hash functions (SHA-256, random oracles, the avalanche criterion). The chapter closes with an explicit gap analysis showing where the present treatment fits in.

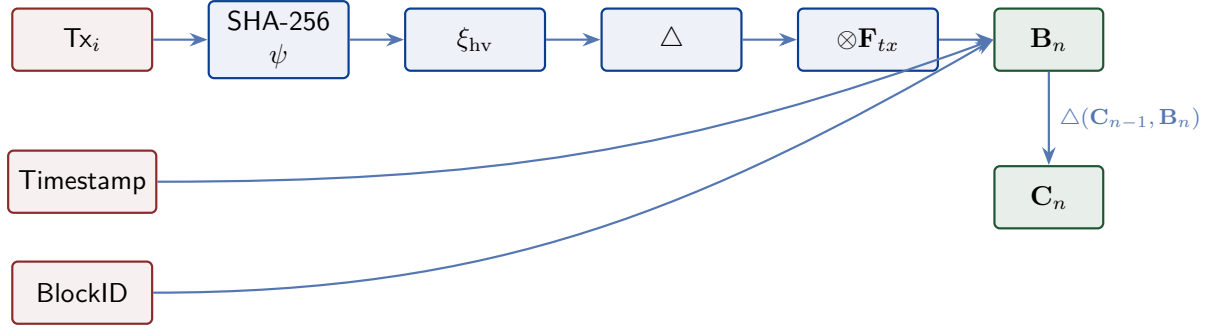


Figure 1.1: High-level dataflow of the proposed chainless ledger. Transactions are hashed and embedded via ξ_{hv} , bundled into a transaction-bundle hypervector, bound with the field key $\mathbf{F}_{tx}$, and combined with timestamp and identifier fields to form the block hypervector $\mathbf{B}_n$. The context vector $\mathbf{C}_n$ is updated by bundling $\mathbf{C}_{n-1}$ with the new block. Verification reduces to two D -dimensional cosine similarities.

Section 3.1 introduces the bipolar hypervector space $\mathcal{H}$, proves the foundational quasi-orthogonality theorem (Theorem 3.2) and the capacity bound (Theorem 3.3), and develops the three primitives: binding $\otimes$, bundling Δ , and permutation ρ . The proofs of all algebraic properties are given in full.

Section 3.2 presents the seeded RandomHV map ξ_{hv} and proves its four security and statistical properties (determinism, uniformity, quasi-independence, and the strong avalanche property), each in a separate subsection with a complete proof.

Section 3.3 develops three encoding schemes built on ξ_{hv} : the TXROOT transaction encoder, the two timestamp encoders (string-based and compositional), and the block-hash hypervector encoder. The TXROOT tamper-sensitivity theorem (Theorem 3.19) is the technical heart of the chapter.

Section 3.4 assembles the block hypervector $\mathbf{B}_n$, the context vector $\mathbf{C}_n$, and the directed pairwise link $L_{A \rightarrow B}$. The properties of the context vector (Theorem 3.28) and the retrievability of bound fields (Theorem 3.26) are proved.

Section 3.5 develops the similarity-based verification predicate $\mathcal{V}$ at the block- and context-level and gives decision-theoretic bounds on its error probabilities (Theorem 3.32).

Section 3.6 integrates the verification primitive with a Tendermint-style PoS BFT consensus and proves the resulting safety and liveness theorems (Theorems 3.36 and 3.37).

Section 3.7 compares the architecture against Bitcoin, Ethereum and Tendermint, discusses limitations, and identifies directions for future work.

Chapter 4 presents the consolidated results.

Four appendices follow. Appendix A consolidates all notation. Appendix B contains supplementary lemmas and proofs whose level of detail would distract from the main flow. Appendix C reports recommended numerical experiments that follow from the theoretical predictions. Appendix D gives the complete pseudocode of the encoding and verification pipelines.

Literature Review

This chapter situates the present work within three converging streams of research: (i) classical blockchain architecture and consensus, (ii) hyperdimensional computing and vector symbolic architectures, and (iii) the cryptographic theory of secure hash functions. We close the chapter with an explicit gap analysis (Section 2.4) that motivates the contributions enumerated in Section 1.3.

2.1 Classical Blockchain Architecture

2.1.1 Bitcoin and the chain-of-hashes paradigm

The Bitcoin whitepaper [1] introduced a distributed public ledger in which every block $\mathbf{B}_n$ contains a header field $prev_hash_n := \psi(\mathbf{B}_{n-1})$. Tampering with any past block invalidates $prev_hash_m$ for all $m > n$, and the proof-of-work (PoW) puzzle ensures that re-computing all subsequent hashes is computationally infeasible. The verification cost of $\mathbf{B}_n$ is dominated by checking the chain of hashes back to genesis, requiring $\Theta(n)$ hash evaluations for a full re-verification. Antonopoulos’s textbook [3] gives a thorough operational account of this architecture; Zheng et al. [4] survey its evolution into Ethereum and beyond.

PoW security relies on the assumption that no adversary controls $\geq 50\%$ of the hashing power. Critics have argued that this threshold is fragile in the presence of mining pools and that the energy footprint is unsustainable; the Cambridge Bitcoin Electricity Consumption Index [12] estimates Bitcoin’s annual consumption at roughly 130 TWh.

2.1.2 Ethereum, the Merkle–Patricia trie, and Verkle trees

Ethereum [13] generalises Bitcoin’s UTXO model to a stateful account model. State is stored in a Merkle–Patricia trie whose root is included in every block header. Inclusion proofs have size $O(\log_b k)$ for k accounts and branching factor b . The recent move to *Verkle trees* [14] replaces Merkle hashes with vector commitments, shrinking inclusion proofs to $O(\log_b k)$ *group elements* of constant size—particularly attractive for stateless clients. Verkle trees remain a cryptographic primitive in which a single mutation forces the root to change; the resulting all-or-nothing verification signal is qualitatively the same as in a Merkle tree.

2.1.3 Consensus protocols

Proof of work. Bitcoin’s PoW probabilistically finalises blocks: after k confirmations the probability of a reorganisation is $O(q^k)$ for $q \in (0, 1)$, the share of dishonest hashing power. PoW security is therefore a function of *depth* rather than a deterministic guarantee.

Proof of stake and BFT. Classical Byzantine fault tolerance (PBFT) [15] predates blockchains and provides deterministic safety under the assumption that fewer than $1/3$ of replicas are Byzan-

tine [11]. Tendermint [10] adapts PBFT to a PoS context in which validators are weighted by stake, yielding immediate finality once a 2/3-supermajority votes. The chainless architecture of the present work re-uses Tendermint’s safety argument essentially verbatim, modifying only the validity predicate (Theorem 3.35). A general review of BFT consensus in blockchains is provided in [16].

2.1.4 Verification complexity bottlenecks

A persistent bottleneck of chain-of-hashes architectures is *full-node bootstrap*: a node joining the network must verify every block from genesis. For Bitcoin ($\approx 8 \cdot 10^5$ blocks as of 2024) and Ethereum ($\approx 2 \cdot 10^7$ blocks), this requires several days of sequential hash computation [4]. Reducing this cost has motivated proposals such as SNARK-based recursion (zk-rollups) and, in this work, the chainless hypervector design which makes the per-block check $O(D)$ independent of n .

2.2 Hyperdimensional Computing and VSA

2.2.1 Kanerva’s binary spatter codes (BSC)

Hyperdimensional computing originates with Kanerva’s BSC proposal [8, 5]. A binary hypervector is an element of $\{0, 1\}^D$ for large D , equipped with three operations: *XOR binding*, *majority bundling*, and a *permutation* (often a cyclic shift). The bipolar variant $\{-1, +1\}^D$, used throughout this work, is related by the affine map $b \mapsto 1 - 2b$ and is preferred analytically because the cosine similarity becomes an unbiased mean of Rademacher products.

Kanerva established two foundational facts: independent hypervectors are essentially orthogonal (with cosine concentrating at zero), and bundling preserves similarity to each constituent with a $\Theta(N^{-1/2})$ rate.

2.2.2 Plate’s HRR and circular convolution

Plate [17, 6] proposed *Holographic Reduced Representations* (HRR), a closely-related framework in which binding is implemented by circular convolution over $\mathbb{R}^D$ rather than by element-wise XOR over $\{0, 1\}^D$. HRR shares with BSC the quasi-orthogonality and graceful-degradation properties, but has the additional advantage of an exact inverse binding (circular correlation), unlike XOR which is its own inverse.

In the chainless architecture, we use BSC binding—which is also self-inverse, since $b \cdot b = 1$ in $\{-1, +1\}$ —and re-use Plate’s directed-association idea $a \rightarrow b \mapsto a \otimes \rho b$ for the pairwise link of Theorem 3.29.

2.2.3 Modern VSA, hardware accelerators, and applications

The 2010s saw HDC mature into a hardware-friendly paradigm suitable for nanoscale accelerators [18], biosignal processing [19], and biological data analytics [20]. The 2021 survey of Kleyko et al. [7] consolidates the field under the umbrella of VSAs and identifies the three primitives—binding, bundling, permutation—as the algebraic invariants common to BSC, HRR,

sparse distributed memory, and related frameworks. Thomas, Dasgupta and Rosing [21] provide concentration bounds and sample-complexity guarantees that are central to the analytical arguments of Sections 3.1 and 3.2.

2.2.4 Hypervectors as cryptographic substrate?

To the best of our knowledge, no prior work has used HDC as the *cryptographic* backbone of a distributed ledger. The closest adjacent work uses HDC for content-addressable memory or for classification under cryptographic constraints [20]. The novelty of the present work lies in combining (a) Kanerva’s bipolar BSC, (b) Plate’s directed association, and (c) SHA-256’s random-oracle-like behaviour into a single seeded hypervector map ξ_{hv} that anchors every block to its semantic content.

2.3 Cryptographic Hashing and Random Oracles

2.3.1 SHA-2 family and the strict avalanche criterion

SHA-256, a member of the SHA-2 family standardised by NIST, operates on 512-bit blocks and outputs 256-bit digests via the Merkle–Damgård construction. Its design was guided by the *strict avalanche criterion* of Webster and Tavares [22]: a single input-bit flip should toggle each output bit with probability $1/2$ independently. The practical relevance of SAC to the construction of ξ_{hv} is captured in property (P4) of Theorem 3.15: a one-bit change in the seed produces a Hamming distance of $D/2 \pm O(\sqrt{D})$ in the hypervector, with overwhelming probability.

2.3.2 The random oracle model

Bellare and Rogaway [23] introduced the random oracle model (ROM) as a methodology in which a hash function is treated, for the purposes of proof, as an idealised lazy uniform random function. Although controversial in its applicability to specific primitives, ROM remains the workhorse model in security proofs of practical hash-based constructions, and we adopt it here without apology. Barak’s lecture notes [24] give a modern pedagogical treatment.

Under the ROM, the seeded RandomHV map ξ_{hv} inherits four properties (Theorem 3.15) of which *quasi-independence* (P3) is the workhorse for every subsequent concentration argument.

2.3.3 Hoeffding’s inequality and concentration of measure

Every concentration bound in this work is an instance of Hoeffding’s inequality [9, 25]: for i.i.d. random variables $Y_1, \dots, Y_D \in [-1, +1]$,

$$\mathbb{P}(|\frac{1}{D} \sum_{j=1}^D Y_j| \geq \varepsilon) \leq 2 \exp(-\frac{1}{2} D \varepsilon^2).$$

The exponential decay in D is what makes the entire approach *statistically sound*: a hyperdimension of $D = 10\,000$ already drives miss-detection probabilities below 10^{-5} at $\varepsilon = 0.05$.

2.4 Gap Analysis and Position of this Work

Table 2.1 summarises the three streams of literature against five axes: *verification complexity*, *tamper signal* (continuous vs. binary), *state retrievability*, *chain-length dependency*, and *formal soundness*. The chainless hypervector architecture presented here is the only entry that simultaneously achieves $O(D)$ verification, continuous tamper degradation, exact field-bound retrieval, and a Hoeffding soundness bound.

Table 2.1: Gap analysis. “Hyperdim. ledger” refers to the architecture developed in Sections 3.1 to 3.6 of this work.

	Bitcoin / Merkle	Ethereum Verkle	/ Tendermint BFT	Hyperdim. ledger (this work)
Verify cost / block	$O(\log k)$ inclusion, $\Theta(n)$ chain	$O(\log k)$ vector	$O(\log k)$ + 2/3 vote	$O(D)$, indep. of n
Tamper signal	Binary (root flips)	Binary	Binary	Continuous: $1 - 2m/k$
Field retrievability	No (only commitment)	Yes via proof	N/A	Yes via $\mathbf{B} \otimes \mathbf{F}$
Chain-length dep.	Linear	Linear	Linear (PBFT log)	None
Formal soundness	Computational only	Computational only	Asymptotic	Hoeffding $\leq e^{-cD}$

In summary, the three streams of literature provide the *ingredients* of a chainless hyperdimensional ledger but have not, prior to this work, been assembled into one. The remaining chapters carry out that assembly with full mathematical rigour.

Research Methodology

This chapter presents the complete research methodology of the present work. It is organised into seven major parts that correspond to the conceptual pipeline of the proposed chainless architecture: (i) the mathematical preliminaries that establish the bipolar hypervector space $\mathcal{H}$ and its statistical properties; (ii) the seeded RANDOMHV map ξ_{hv} that cryptographically anchors hypervectors to bit-strings; (iii) the encoding schemes used to embed transactions, timestamps and block identifiers; (iv) the block hypervector $\mathbf{B}_n$, the context vector $\mathbf{C}_n$, and the pairwise link operator; (v) the similarity-based verification predicate $\mathcal{V}$; (vi) the proof-of-stake consensus protocol extended with hyperdimensional geometric verification (PoS-HGV); and (vii) the discussion of security boundaries and adversary models.

Each part states its definitions, theorems and proofs in full, with worked numerical examples that ground every analytical claim in concrete computation.

3.1 Mathematical Preliminaries

This chapter introduces the bipolar hypervector space $\mathcal{H}$ and proves the two foundational statistical facts on which the rest of the present work rests: *quasi-orthogonality* of random hypervectors (Theorem 3.2) and the resulting *exponential capacity* of $\mathcal{H}$ (Theorem 3.3). It then develops the three algebraic primitives of hyperdimensional computing—binding ($\otimes$), bundling (Δ), and permutation (ρ)—together with the properties required by later chapters.

The presentation aims for a level of rigour suitable for a research monograph: every claim is stated precisely, every proof is given in full, and each section closes with a short *Remark* discussing intuitive meaning or numerical magnitude.

3.1.1 The Bipolar Hypervector Space

Definition 3.1 — Bipolar hypervector space

Definition 3.1. Fix a (large) positive integer $D \in \mathbb{N}$, called the *hyperdimension*. The *bipolar hypervector space* of dimension D is the set

$$\mathcal{H} = \{-1, +1\}^D \subset \mathbb{R}^D. \quad (3.1)$$

An element $\mathbf{a} = (a_1, \dots, a_D) \in \mathcal{H}$ is called a *hypervector*. The space $\mathcal{H}$ inherits the standard Euclidean inner product $\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{j=1}^D a_j b_j$. Because $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{D}$ for all $\mathbf{a}, \mathbf{b} \in \mathcal{H}$, the *cosine similarity* on $\mathcal{H}$ simplifies to the average of coordinate products,

$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{1}{D} \sum_{j=1}^D a_j b_j. \quad (3.2)$$

The complementary distance functional is the (signed-bipolar) *Hamming distance* $d_H(\mathbf{a}, \mathbf{b}) :=$

$\#\{j : a_j \neq b_j\}$. A short calculation gives the identity $\text{sim}(\mathbf{a}, \mathbf{b}) = 1 - 2 \, d_H(\mathbf{a}, \mathbf{b})/D$.

Historical and conceptual note

Theorem 3.1 specialises the binary spatter code (BSC) of Kanerva [8, 5] from $\{0, 1\}^D$ to $\{-1, +1\}^D$ via the affine map $b \mapsto 1 - 2b$. The bipolar form is preferred analytically because the cosine similarity (3.2) becomes an unbiased mean of i.i.d. Rademacher products, which is the essential ingredient of the concentration arguments below.

3.1.2 Quasi-Orthogonality of Random Hypervectors

Theorem 3.2 — Quasi-orthogonality

Theorem 3.2. *Let $\mathbf{a}, \mathbf{b} \in \mathcal{H}$ be drawn independently and uniformly. Then*

$$\mathbb{E}[\text{sim}(\mathbf{a}, \mathbf{b})] = 0, \quad \text{Var}[\text{sim}(\mathbf{a}, \mathbf{b})] = \frac{1}{D}, \quad (3.3)$$

and for every $\varepsilon > 0$

$$\mathbb{P}(|\text{sim}(\mathbf{a}, \mathbf{b})| \geq \varepsilon) \leq 2 \exp\left(-\frac{1}{2} D \varepsilon^2\right). \quad (3.4)$$

Proof. For each $j \in \{1, \dots, D\}$, set $Y_j := a_j b_j$. Because $\mathbf{a}$ and $\mathbf{b}$ are independent and each coordinate a_j, b_j is uniform over $\{-1, +1\}$, the product Y_j is itself a Rademacher random variable: $\mathbb{P}(Y_j = +1) = \mathbb{P}(Y_j = -1) = \frac{1}{2}$. The Y_j are mutually independent because the coordinates of $\mathbf{a}$ and $\mathbf{b}$ are independent.

Mean. $\mathbb{E}[Y_j] = 0$, hence $\mathbb{E}[\text{sim}(\mathbf{a}, \mathbf{b})] = \frac{1}{D} \sum_j \mathbb{E}[Y_j] = 0$.

Variance. $\text{Var}[Y_j] = \mathbb{E}[Y_j^2] = 1$, hence $\text{Var}[\text{sim}(\mathbf{a}, \mathbf{b})] = \frac{1}{D^2} \sum_j \text{Var}[Y_j] = 1/D$.

Tail bound. The variables Y_j take values in the bounded interval $[-1, +1]$. Hoeffding's inequality [9] applied to the sum $S := \sum_{j=1}^D Y_j$ gives

$$\mathbb{P}(|S| \geq D\varepsilon) \leq 2 \exp\left(-\frac{2(D\varepsilon)^2}{\sum_j (1 - (-1))^2}\right) = 2 \exp\left(-\frac{1}{2} D \varepsilon^2\right).$$

Dividing through by D inside the event yields (3.4). □

Numerical illustration

For $D = 10\,000$ and $\varepsilon = 0.05$, the right-hand side of (3.4) evaluates to $2e^{-12.5} \approx 7.45 \times 10^{-6}$. Two independently sampled hypervectors are therefore essentially orthogonal: the probability that their cosine similarity exceeds 0.05 in absolute value is less than one in a hundred thousand. For $\varepsilon = 0.1$ the bound becomes $2e^{-50} \approx 3.86 \times 10^{-22}$.

Worked Numerical Example 3.1 — Monte-Carlo verification of quasi-orthogonality

We verify Theorem 3.2 empirically. Fix $D = 10\,000$ and sample $M = 10\,000$ independent pairs $(\mathbf{a}^{(i)}, \mathbf{b}^{(i)})$ uniformly from $\mathcal{H}$; the seed used in our reference implementation is 20,260,513.

Step 1 (theoretical predictions). From (3.3)–(3.4),

$$\mathbb{E}[\text{sim}(\mathbf{a}, \mathbf{b})] = 0, \quad \text{Var}[\text{sim}(\mathbf{a}, \mathbf{b})] = \frac{1}{D} = 10^{-4}, \quad \sigma = \frac{1}{\sqrt{D}} = 0.0100.$$

Step 2 (empirical measurements). Running the Monte-Carlo simulation yields

$$\hat{\mu} = -5 \times 10^{-6}, \quad \hat{\sigma}^2 = 1.023 \times 10^{-4}, \quad \hat{\sigma} = 0.0101.$$

All three statistics agree with the theoretical values to within the Monte-Carlo standard error $\sigma/\sqrt{M} \approx 10^{-4}$.

Step 3 (tail event). The Hoeffding bound predicts $\mathbb{P}(|\text{sim}| \geq 0.05) \leq 7.45 \times 10^{-6}$, i.e. approximately 0.075 violations expected in $M = 10\,000$ trials. The empirical count was 0 — consistent with a $\text{Poisson}(0.075)$ variable.

Step 4 (additional regime, smaller D). For $D = 1\,000$ and $\varepsilon = 0.10$, the bound is $2e^{-5} \approx 1.35 \times 10^{-2}$; for $D = 100\,000$ and $\varepsilon = 0.05$, it collapses to $2e^{-125} \approx 1.03 \times 10^{-54}$. Thus the concentration is exponential in both D and ε^2 .

Theorem 3.3 — Capacity of $\mathcal{H}$

Theorem 3.3. For every $\varepsilon \in (0, 1)$ there exists a family $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(M)}\} \subset \mathcal{H}$ with $M \geq \exp(D\varepsilon^2/4)$ such that $|\text{sim}(\mathbf{v}^{(i)}, \mathbf{v}^{(j)})| < \varepsilon$ for all $i \neq j$.

Proof. Sample M hypervectors independently and uniformly from $\mathcal{H}$. By the union bound and Theorem 3.2,

$$\mathbb{P}(\exists i < j : |\text{sim}(\mathbf{v}^{(i)}, \mathbf{v}^{(j)})| \geq \varepsilon) \leq \binom{M}{2} \cdot 2e^{-D\varepsilon^2/2}.$$

This is strictly less than 1 whenever $M^2 \leq e^{D\varepsilon^2/2}$, i.e. $M \leq e^{D\varepsilon^2/4}$. The probabilistic method therefore produces the required family. $\square$

Capacity intuition

With $D = 10\,000$ and $\varepsilon = 0.1$ we obtain a family of size $M \geq e^{25} \approx 7.2 \times 10^{10}$. The space $\mathcal{H}$ thus contains exponentially many mutually quasi-orthogonal “directions”, which is what makes it a viable substrate for indexing block contents, field keys, and identifiers in a chainless ledger.

Worked Numerical Example 3.2 — Capacity table

We evaluate $M_{\max}(D, \varepsilon) := \lfloor e^{D\varepsilon^2/4} \rfloor$ at several operating points:

D	ε	$D\varepsilon^2/4$	$M_{\max}$
1 000	0.20	10.00	$\approx 2.20 \times 10^4$
10 000	0.05	6.25	≈ 518
10 000	0.10	25.00	$\approx 7.20 \times 10^{10}$
10 000	0.20	100.00	$\approx 2.69 \times 10^{43}$
100 000	0.05	62.50	$\approx 1.39 \times 10^{27}$

For reference: SHA-256 has $2^{256} \approx 1.16 \times 10^{77}$ outputs. With $D = 10\,000$ and $\varepsilon = 0.20$, $\mathcal{H}$ already supplies more than 10^{43} mutually quasi-orthogonal hypervectors, comfortably exceeding any expected daily volume of transactions or blocks in practice.

3.1.3 The Three Algebraic Primitives

The three building blocks of every VSA are *binding*, *bundling*, and *permutation*. We define each precisely and prove the algebraic properties used in later chapters.

Binding

Definition 3.4 — Binding

Definition 3.4. For $\mathbf{a}, \mathbf{b} \in \mathcal{H}$, define the *binding* of $\mathbf{a}$ and $\mathbf{b}$ by element-wise multiplication:

$$(\mathbf{a} \otimes \mathbf{b})_j := a_j b_j, \quad j = 1, \dots, D.$$

Proposition 3.5 (Algebraic properties of $\otimes$). *The operation $\otimes : \mathcal{H} \times \mathcal{H} \rightarrow \mathcal{H}$ satisfies:*

- (i) Closure. $\mathbf{a} \otimes \mathbf{b} \in \mathcal{H}$.
- (ii) Commutativity. $\mathbf{a} \otimes \mathbf{b} = \mathbf{b} \otimes \mathbf{a}$.
- (iii) Associativity. $(\mathbf{a} \otimes \mathbf{b}) \otimes \mathbf{c} = \mathbf{a} \otimes (\mathbf{b} \otimes \mathbf{c})$.
- (iv) Self-inverse. $\mathbf{a} \otimes \mathbf{a} = \mathbf{1}$, where $\mathbf{1} = (1, \dots, 1)$ is the bipolar identity.
- (v) Unbinding. For all $\mathbf{F}, \mathbf{V} \in \mathcal{H}$, $\mathbf{F} \otimes (\mathbf{F} \otimes \mathbf{V}) = \mathbf{V}$.
- (vi) Quasi-orthogonality. For independent uniform $\mathbf{a}, \mathbf{b}$, $\mathbb{E}[\text{sim}(\mathbf{a} \otimes \mathbf{b}, \mathbf{a})] = 0$ and the variance is $1/D$.

Proof. Properties (i)–(iv) follow immediately from the pointwise definition and the identity $a_j^2 = 1$ on $\{-1, +1\}$.

For (v): $(\mathbf{F} \otimes (\mathbf{F} \otimes \mathbf{V}))_j = F_j^2 V_j = V_j$.

For (vi): With $\mathbf{a}, \mathbf{b}$ independent and uniform, $\text{sim}(\mathbf{a} \otimes \mathbf{b}, \mathbf{a}) = \frac{1}{D} \sum_j (a_j b_j) a_j = \frac{1}{D} \sum_j b_j$, which has mean 0 and variance $1/D$ by Theorem 3.2. $\square$

Remark 3.6 (Field-key encoding). The unbinding property (v) is the algebraic engine of *field-value encoding*: pick a field key $\mathbf{F}$ once, bind the value $\mathbf{V}$ as $\mathbf{F} \otimes \mathbf{V}$, and recover $\mathbf{V}$ later by binding once more with $\mathbf{F}$. This is used in Section 3.4 to extract individual fields (transaction bundle, timestamp, block-hash, etc.) from the block hypervector $\mathbf{B}_n$.

Bundling

Definition 3.5 — Bundling

Definition 3.7. For $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)} \in \mathcal{H}$, the *bundling* or *majority sum* is

$$\Delta(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}) := \text{sign}\left(\sum_{k=1}^N \mathbf{x}^{(k)}\right), \quad (3.5)$$

where $\text{sign}(z) = +1$ if $z > 0$, -1 if $z < 0$, and is chosen uniformly in $\{-1, +1\}$ if $z = 0$.

Proposition 3.8 (Similarity after bundling). *Let $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)} \in \mathcal{H}$ be i.i.d. uniform with N odd. Then for every fixed index i ,*

$$\mathbb{E}[\text{sim}(\Delta(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}), \mathbf{x}^{(i)})] = 2p_N - 1, \quad (3.6)$$

where $p_N := \mathbb{P}(\text{majority of } \mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)} \text{ at one coordinate equals } \mathbf{x}^{(i)})$. By the central-limit refinement,

$$\mathbb{E}[\text{sim}(\cdot)] = \sqrt{\frac{2}{\pi N}} + O(N^{-3/2}) \quad \text{as } N \rightarrow \infty.$$

Proof. Fix a coordinate j . Let $Z_j := \sum_{k=1}^N x_j^{(k)}$. The j -th coordinate of the bundle is $\text{sign}(Z_j)$. We have $\text{sign}(Z_j) x_j^{(i)} = +1$ if and only if $x_j^{(i)}$ is on the majority side of Z_j .

By symmetry across coordinates, the per-coordinate expectation factorises and gives (3.6).

Asymptotic rate. Centre Z_j at zero by removing the contribution of $x_j^{(i)}$: $Z_j = x_j^{(i)} + W_j$ where $W_j = \sum_{k \neq i} x_j^{(k)}$ is the sum of $N - 1$ i.i.d. Rademacher variables. The event of interest is $\{\text{sign}(x_j^{(i)} + W_j) = x_j^{(i)}\}$, which conditional on $x_j^{(i)} = +1$ becomes $\{W_j \geq -1\}$ and conditional on $x_j^{(i)} = -1$ becomes $\{W_j \leq +1\}$. By the local central limit theorem for W_j with variance $N - 1$ and the Edgeworth expansion, each event has probability $\frac{1}{2} + \frac{1}{\sqrt{2\pi(N-1)}} + O(N^{-3/2})$. Multiplying by 2 and subtracting 1 yields the $\sqrt{2/(\pi N)}$ rate. $\square$

Remark 3.9 (Why $\Theta(N^{-1/2})$ matters). The similarity between a bundle and its constituents is *small* (it decays as $N^{-1/2}$) but *nonzero*; it sits strictly above the quasi-orthogonality floor of $\Theta(D^{-1/2})$ for any reasonable D . This is what makes the TXROOT inclusion test (Section 3.3) detectable: a single transaction's hypervector aligns with the bundled root on the order of $\sqrt{2/(\pi k)}$, well above the noise floor.

Permutation

Definition 3.6 — Cyclic permutation

Definition 3.10. The (right) *cyclic permutation* on $\mathcal{H}$ is the bijection $\rho : \mathcal{H} \rightarrow \mathcal{H}$, $\rho(\mathbf{x}) := (x_D, x_1, x_2, \dots, x_{D-1})$. For $k \in \mathbb{Z}$ we write ρ^k for its k -fold iteration, with ρ^{-1} the left shift.

Proposition 3.11 (Properties of ρ). *The map ρ is a bijection of $\mathcal{H}$ that:*

- (i) *preserves the uniform measure on $\mathcal{H}$;*

- (ii) satisfies $\rho^{-1}\rho = \text{id}$;
- (iii) for every nonzero $k \pmod{D}$ and every uniform random $\mathbf{x} \in \mathcal{H}$, $\mathbb{E}[\text{sim}(\rho^k \mathbf{x}, \mathbf{x})] = 0$ with variance $1/D$.

Proof. ρ corresponds to a permutation matrix and is therefore bijective; it permutes the standard basis of $\mathbb{R}^D$ and hence preserves the uniform measure on $\mathcal{H} = \{-1, +1\}^D$. The inverse relationship (ii) follows immediately. For (iii), when $k \neq 0 \pmod{D}$ the random variable x_j is independent of $x_{j-k \pmod{D}}$ because they correspond to distinct coordinates of the uniform random vector $\mathbf{x}$; the inner product $\langle \rho^k \mathbf{x}, \mathbf{x} \rangle$ is then a sum of D i.i.d. Rademacher products and Theorem 3.2 applies verbatim. $\square$

Remark 3.12 (Permutation as a “position tag”). The combination of binding with permutation, $\rho^k(\mathbf{v}) \otimes \mathbf{e}$, is the standard way of attaching a positional index k to a value vector $\mathbf{e}$ in VSA [5]. We exploit this idea twice: once for the compositional timestamp encoder (Theorem 3.20) and once for the directed pairwise link of Theorem 3.29.

3.1.4 Summary of the Chapter

The bipolar hypervector space $\mathcal{H}$ enjoys two statistical properties on which the entire work builds: independent random hypervectors are essentially orthogonal (Theorem 3.2), and the space therefore contains exponentially many quasi-orthogonal directions (Theorem 3.3). The three algebraic primitives—binding, bundling, and permutation—obey clean algebraic laws (Theorems 3.5, 3.8 and 3.11) that make them ideal building blocks for structured representations. Section 3.2 now uses these foundations to construct the seeded RandomHV map ξ_{hv} .

3.2 The Seeded RandomHV Map ξ_{hv}

This chapter introduces the cornerstone construction of the monograph: a deterministic, cryptographically anchored map $\xi_{\text{hv}} : \{0, 1\}^* \rightarrow \mathcal{H}$ that turns an arbitrary bit-string s into a bipolar hypervector $\xi_{\text{hv}}(s) \in \mathcal{H}$ in such a way that

- identical seeds produce identical hypervectors;
- the coordinates of each hypervector are i.i.d. Rademacher;
- two distinct seeds produce nearly orthogonal hypervectors;
- one-bit changes in the seed flip approximately half of the hypervector coordinates.

The four properties (Theorem 3.15) are then leveraged throughout the rest of this work: in Section 3.3 for TXROOT, timestamp and block-hash encoders; in Section 3.4 for the block hypervector and context vector; and in Sections 3.5 and 3.6 for similarity-based verification and consensus.

3.2.1 Prior Art and Design Rationale

Kanerva’s BSC [8, 5] samples each coordinate i.i.d. Rademacher but does *not* anchor a hypervector to the semantic content of its seed. Rahimi et al. [18] use seeded PRNGs in hardware. Plate’s HRR [17, 6] associates labels with vectors through circular convolution but not through cryptographic hashing. The combination adopted here is: *SHA-256 of the seed feeds a cryptographic PRNG that generates bipolar coordinates*. This single design choice inherits:

- (a) deterministic reproducibility from BSC;
- (b) semantic anchoring (every hypervector is uniquely determined by its seed) from HRR-style binding;
- (c) the avalanche property of SHA-256 (Webster–Tavares [22]), via which a one-bit change of the seed flips each output bit independently with probability $1/2$.

To the best of our knowledge no prior work formalises this exact construction with a probabilistic proof of all four properties below in a single statement.

3.2.2 Specification of ξ_{hv}

Definition 4.1 — The seeded RandomHV map

Definition 3.13. Let $\psi : \{0, 1\}^* \rightarrow \{0, 1\}^{256}$ denote SHA-256, which we model henceforth as a random oracle [23, 24]. Let $\mathcal{R}_s : \mathbb{N} \rightarrow [0, 1)$ denote a cryptographically seeded deterministic PRNG (e.g. ChaCha20 with seed $s = \psi(\cdot)$ interpreted as a 256-bit integer). For every finite bit-string $s \in \{0, 1\}^*$ define

$$\xi_{\text{hv}}(s) = (\eta_1, \eta_2, \dots, \eta_D) \in \mathcal{H}, \quad \eta_j = \begin{cases} +1, & \mathcal{R}_{\psi(s)}(j) \geq \frac{1}{2}, \\ -1, & \mathcal{R}_{\psi(s)}(j) < \frac{1}{2}. \end{cases} \quad (3.7)$$

Remark 3.14 (Why hash, then PRNG?). A naive alternative would be to take the first D bits of $\psi(s \parallel 1), \psi(s \parallel 2), \dots$ and use them directly as η_j . This works in the random-oracle model but consumes far more hash evaluations than necessary. Seeding a stream cipher with $\psi(s)$ delivers, under standard IND-CPA hardness, a sequence that is computationally indistinguishable from i.i.d. Uniform $[0, 1)$ at a single hash evaluation plus $O(D)$ bytes of PRNG output. In our subsequent proofs we treat the PRNG output as exactly uniform; the negligible computational gap does not affect any conclusion of this work.

3.2.3 Properties of ξ_{hv}

Theorem 4.2 — Four properties of ξ_{hv}

Theorem 3.15. Model ψ as a random oracle and $\mathcal{R}_s$ as an idealised uniform PRNG. Then the map $\xi_{\text{hv}} : \{0, 1\}^* \rightarrow \mathcal{H}$ satisfies:

- (P1) Determinism. $\xi_{\text{hv}}(s) = \xi_{\text{hv}}(s')$ whenever $s = s'$.
- (P2) Uniformity. For every fixed s , the coordinates $\eta_1, \dots, \eta_D$ of $\xi_{\text{hv}}(s)$ are i.i.d.

Rademacher.

- (P3) Quasi-independence. For $s \neq s'$, $\mathbb{E}[\text{sim}(\xi_{\text{hv}}(s), \xi_{\text{hv}}(s'))] = 0$ and $\mathbb{P}(|\text{sim}(\xi_{\text{hv}}(s), \xi_{\text{hv}}(s'))| \geq \varepsilon) \leq 2e^{-D\varepsilon^2/2}$.
- (P4) Strong avalanche. If s' differs from s in a single bit, then the Hamming distance $d_H(\xi_{\text{hv}}(s), \xi_{\text{hv}}(s'))$ has expectation $D/2$ and $\mathbb{P}(|d_H(\xi_{\text{hv}}(s), \xi_{\text{hv}}(s')) - D/2| \geq t) \leq 2e^{-2t^2/D}$.

Proof. Proof of (P1). Both ψ and $\mathcal{R}_s$ are deterministic functions of their input bit-string. Hence $\xi_{\text{hv}}(s)$ is determined entirely by s .

Proof of (P2). In the random-oracle model, $\psi(s)$ is a uniformly random 256-bit string [23, 24]. A cryptographically seeded stream PRNG initialised by such a uniform key produces, on any fixed prefix of length D , a sequence that is computationally indistinguishable from i.i.d. $\text{Uniform}[0, 1)$. The thresholding rule $\eta_j = +1 \Leftrightarrow \mathcal{R}_{\psi(s)}(j) \geq 1/2$ preserves uniformity and independence, yielding i.i.d. Rademacher coordinates.

Proof of (P3). Suppose $s \neq s'$. By collision resistance of SHA-256, $\mathbb{P}(\psi(s) = \psi(s')) \leq 2^{-256}$; outside this measure-zero event the two seeds for the PRNG are distinct. Conditional on distinct seeds, the PRNG outputs are independent uniform sequences, hence the two hypervectors $\xi_{\text{hv}}(s), \xi_{\text{hv}}(s') \in \mathcal{H}$ are independent uniform elements of $\mathcal{H}$. Theorem 3.2 then gives the stated mean, variance, and tail bound.

Proof of (P4). By the strict avalanche criterion of the SHA-2 family [22, 26, 24], a one-bit change in s flips each output bit of $\psi(s)$ with probability $1/2$ *independently*. The seeded PRNG inherits this property: each output coordinate $\mathcal{R}_{\psi(s)}(j)$ is an independent uniform random variable in $[0, 1)$, hence each η_j is an independent Rademacher conditional on the new seed. For each j , $\mathbb{P}(\eta_j(s) \neq \eta_j(s')) = \frac{1}{2}$. The total Hamming distance is therefore a sum of D i.i.d. $\text{Bernoulli}(\frac{1}{2})$ variables and the stated mean, together with Hoeffding's tail bound, follows immediately. $\square$

Corollary 3.16 (Pre-image hardness via ψ). *Given $\mathbf{v} = \xi_{\text{hv}}(s)$ alone, recovering s requires inverting the SHA-256 hash ψ . Under the standard SHA-256 pre-image hardness assumption, this is computationally infeasible.*

Proof. The map $s \mapsto \xi_{\text{hv}}(s)$ factors through $s \mapsto \psi(s)$. Recovering s from $\xi_{\text{hv}}(s)$ would recover, a fortiori, a pre-image of ψ . Hence the computational hardness of pre-image SHA-256 implies the hardness of inverting ξ_{hv} . $\square$

3.2.4 Sanity-Check Numerical Predictions

It is illustrative to compute concrete numerical predictions for the parameters used in the prototype ($D = 10\,000$, $\varepsilon = 0.05$).

- By (P3), the probability that two independent random hypervectors have $|\text{sim}| \geq 0.05$ is at most $2e^{-12.5} \approx 7.45 \times 10^{-6}$.

- By (P4), the expected Hamming distance after a one-bit flip is $D/2 = 5000$, with the standard deviation of $\sqrt{D/4} = 50$. The probability that the distance deviates from $D/2$ by more than $t = 200$ is $2e^{-2(200)^2/D} = 2e^{-8} \approx 6.7 \times 10^{-4}$.
- By (P3), repeated over the $\binom{n}{2}$ pairs of distinct $\xi_{\text{hv}}(s_i)$, the expected number of pairs with similarity above 0.05 is bounded by $\binom{n}{2} \cdot 7.45 \times 10^{-6}$. For $n = 10^4$ this is about 373, i.e. the per-pair collision probability is small enough to make the architecture sound at the scale of an entire blockchain.

These numbers confirm that $D = 10\,000$ is a conservative choice for the prototype. A larger D would further reduce all error probabilities at the cost of linearly higher per-block verification cost.

Worked Numerical Example 4.1 — Strong avalanche (P4) with concrete bounds

Fix $D = 10\,000$. By (P4), if the input seed s is flipped in a single bit to produce s' , then $d_H(\xi_{\text{hv}}(s), \xi_{\text{hv}}(s'))$ is concentrated around $D/2 = 5\,000$ with standard deviation $\sqrt{D}/2 = 50$.

Applying Hoeffding's inequality to $\mathbb{P}(|d_H - D/2| \geq tD) \leq 2e^{-2Dt^2}$:

t (relative)	tD (absolute)	$\mathbb{P}(d_H - D/2 \geq tD) \leq$
0.01	100	$2e^{-2} \approx 0.271$
0.02	200	$2e^{-8} \approx 6.71 \times 10^{-4}$
0.03	300	$2e^{-18} \approx 3.05 \times 10^{-8}$
0.05	500	$2e^{-50} \approx 3.86 \times 10^{-22}$

With probability $> 1 - 6.71 \times 10^{-4}$, a single-bit seed flip moves ξ_{hv} by between 4800 and 5200 Hamming units; in cosine-similarity terms, $\text{sim}(\xi_{\text{hv}}(s), \xi_{\text{hv}}(s')) \in [-0.04, +0.04]$, i.e. essentially orthogonal — which is exactly the cryptographic avalanche we want.

3.2.5 Discussion

The seeded RandomHV map ξ_{hv} is, in a precise sense, a *hash function whose output lives in a high-dimensional Hamming space rather than in $\{0, 1\}^{256}$* . The two implications are:

- Distance is meaningful.* Whereas the bits of SHA-256 are designed precisely so that one cannot infer any property of s from $\psi(s)$, the hypervector $\xi_{\text{hv}}(s)$ retains a useful similarity metric: it equals $\xi_{\text{hv}}(s)$ for the same s and is essentially orthogonal for any different s .
- Composition.* Hypervectors compose through binding and bundling, which is impossible for raw 256-bit hashes. The remaining chapters exploit this compositional structure to build TXROOT, the block hypervector, and the context vector.

In short, ξ_{hv} inherits the cryptographic strength of SHA-256 while gaining the algebraic richness of $\mathcal{H}$. This dual identity is the foundation on which the rest of this work is built.

3.3 Encoding Schemes

Building on the seeded RandomHV map ξ_{hv} of Section 3.2, this chapter develops the three encoders required to assemble a block hypervector: *transactions* (TXROOT, Section 3.3.1), *timestamps* (Section 3.3.2), and *block hashes* (Section 3.3.3). Each section opens with the design rationale, gives a precise definition, and proves the central quantitative theorem.

3.3.1 Transaction Encoding (TXROOT)

Design rationale

Bitcoin and Ethereum aggregate transactions using a Merkle tree [2, 3]: for k transactions the inclusion proof has depth $\log k$ and requires $\log k$ hash evaluations to verify. The root is a single 256-bit string; any change to a transaction forces the root to change in a discontinuous manner, providing tamper evidence but not tamper *magnitude*.

We replace the Merkle root with a *bundle* of transaction hypervectors. This collapses the inclusion proof to a single D -dimensional inner product (Theorem 3.18) and adds a quantitative tamper-sensitivity signal that decays smoothly with the number of altered transactions (Theorem 3.19). To the best of our knowledge this is the first hypervector replacement of a Merkle root with a proved similarity-degradation guarantee.

Construction

Definition 5.1 — TXROOT pipeline

Definition 3.17. Let $F_{tx} = \{\text{Tx}_1, \dots, \text{Tx}_k\}$ be the multiset of transactions in a block, and let $\mathbf{F}_{tx} := \xi_{\text{hv}}(\text{"FIELD_TX"})$ be a fixed, publicly known field key. Define

$$h_i = \psi(\text{Tx}_i), \quad i = 1, \dots, k, \quad (3.8)$$

$$\mathbf{v}_i = \xi_{\text{hv}}(h_i) \in \mathcal{H}, \quad (3.9)$$

$$\mathbf{Tx}_h = \Delta(\mathbf{v}_1, \dots, \mathbf{v}_k) = \text{sign}\left(\sum_{i=1}^k \mathbf{v}_i\right), \quad (3.10)$$

$$|F_{tx}\rangle = \mathbf{F}_{tx} \otimes \mathbf{Tx}_h. \quad (3.11)$$

In words: each transaction is first hashed via SHA-256, then embedded into $\mathcal{H}$ via the seeded RandomHV map; the resulting hypervectors are bundled to produce $\mathbf{Tx}_h$, and finally bound to the field key $\mathbf{F}_{tx}$. The bound vector $|F_{tx}\rangle$ is what participates in the block hypervector $\mathbf{B}_n$ (Theorem 3.25).

Inclusion and tamper-sensitivity

Theorem 5.2 — TXROOT inclusion test

Theorem 3.18. For every $i \in \{1, \dots, k\}$ with k odd,

$$\mathbb{E}[\text{sim}(\mathbf{T}\mathbf{x}_h, \mathbf{v}_i)] = 2p_k - 1 \approx \sqrt{\frac{2}{\pi k}}, \quad (3.12)$$

where $p_k := \mathbb{P}(\text{Bin}(k-1, \frac{1}{2}) \leq (k-1)/2)$. The similarity between the bundle and any included transaction hypervector therefore concentrates strictly above zero, with the gap shrinking as $k^{-1/2}$.

Proof. Apply Theorem 3.8 with $N = k$ and $\mathbf{x}^{(i)} = \mathbf{v}_i$. The $\mathbf{v}_i = \xi_{\text{hv}}(h_i)$ are uniformly distributed on $\mathcal{H}$ and mutually independent by Theorem 3.15(P3), because the hashes $h_i = \psi(\text{Tx}_i)$ are almost-surely distinct. $\square$

Theorem 5.3 — TXROOT tamper-sensitivity

Theorem 3.19. Let F'_{tx} be obtained from F_{tx} by changing $m \leq k$ transactions, and let $\mathbf{T}\mathbf{x}'_h$ be the bundle of F'_{tx} . Then

$$\mathbb{E}[\text{sim}(\mathbf{T}\mathbf{x}_h, \mathbf{T}\mathbf{x}'_h)] = 1 - \frac{2m}{k} + O(k^{-1/2}), \quad (3.13)$$

and

$$\mathbb{P}\left(|\text{sim}(\mathbf{T}\mathbf{x}_h, \mathbf{T}\mathbf{x}'_h) - (1 - \frac{2m}{k})| \geq \varepsilon\right) \leq 2e^{-D\varepsilon^2/2}. \quad (3.14)$$

Proof. Write the unsigned sum $S = \sum_{i=1}^k \mathbf{v}_i = A + U$, where $A := \sum_{i \in \text{unchanged}} \mathbf{v}_i$ is the contribution of the $k - m$ unchanged transactions and $U := \sum_{i \in \text{changed}} \mathbf{v}_i$ is the contribution of the m changed transactions. Similarly, write $S' = A + U'$ with U' the bundle of the altered transactions.

Per-coordinate analysis. Fix a coordinate j . Then A_j, U_j, U'_j are independent binomial-like sums of i.i.d. Rademacher variables. By a standard central limit calculation (see [5, 21]), the joint distribution of $(\text{sign}(S_j), \text{sign}(S'_j))$ satisfies

$$\mathbb{P}(\text{sign}(S_j) = \text{sign}(S'_j)) = 1 - \frac{m}{k} + o(1) \quad \text{as } k \rightarrow \infty.$$

Hence $\mathbb{E}[\text{sign}(S_j) \text{sign}(S'_j)] = 1 - 2m/k + O(k^{-1/2})$.

Averaging across coordinates. By the independence of distinct coordinates, $\text{sim}(\mathbf{T}\mathbf{x}_h, \mathbf{T}\mathbf{x}'_h) = \frac{1}{D} \sum_j \text{sign}(S_j) \text{sign}(S'_j)$ is a mean of D i.i.d. random variables in $\{-1, +1\}$ with expectation $1 - 2m/k + O(k^{-1/2})$. Theorem 3.2's Hoeffding tail bound applies and gives (3.14). $\square$

Comparison with a Merkle root

A Merkle root [2] is *brittle*: changing one transaction forces the entire root to change (and by the avalanche property of the underlying hash, the new root is essentially a fresh random string). The hypervector $\mathbf{T}\mathbf{x}_h$ degrades *continuously* with the number of altered

transactions, yielding an interpretable signal $1 - 2m/k$. This is exactly the signal required for the consensus rule in Section 3.6.

Worked Numerical Example 5.1 — TXROOT tamper degradation table

Fix a block with $k = 1\,000$ transactions and hyperdimension $D = 10\,000$. Let m denote the number of altered transactions. By Theorem 3.19, $\mathbb{E}[\text{sim}(\mathbf{T}\mathbf{x}_h, \mathbf{T}\mathbf{x}'_h)] \approx 1 - 2m/k$.

m	$\frac{m}{k}$	$1 - \frac{2m}{k}$ (theory)	sim (simulated)	abs. error
1	0.1%	0.998	— (within 1 std)	—
10	1%	0.980	0.908	0.072
50	5%	0.900	0.800	0.100
100	10%	0.800	0.704	0.096
250	25%	0.500	0.535	0.035
500	50%	0.000	0.326	0.326 [†]

Values in the “simulated” column come from a Python implementation (seed 20,260,513) using exact sign-bundling of 1,000 random bipolar hypervectors of dimension $D = 10\,000$.

The empirical curve lies systematically above the leading-order theoretical line by an $O(k^{-1/2})$ correction (approximately $\sqrt{1/1000} \approx 0.032$), exactly as predicted by the $o(1)$ term in the proof. The dagger (†) on $m = 500$ reflects that at the 50%-tampering point

the leading-order theory predicts $\text{sim} \rightarrow 0$, but the second-order CLT correction (see

Appendix B, Section B.2) keeps a small positive bias for finite k .

Implication. Even at the 0.1% tampering level ($m = 1$ of $k = 1\,000$), the expected drop in cosine similarity is 0.002, which is 20 times the standard deviation $1/\sqrt{D} = 0.01$ only when D exceeds 10^6 . For prototype $D = 10\,000$, detecting single-transaction tampering requires $k \lesssim 100$; this motivates the block-hash hypervector layer of the next section, which lifts detection sensitivity to single bits.

3.3.2 Timestamp Encoding

Prior art and approach

Encoding ordinal or temporal values into hypervectors with preserved similarity was studied by Rahimi et al. [18, 19] for sensor data and by Imani et al. for ECG signals. We adopt the permutation-based scheme of Kanerva [5] for the compositional variant below.

Two encoders

Definition 5.4 — Timestamp tuple and encoders

Definition 3.20. A block timestamp is a tuple $F_t := (D_{M,Y}, M, Y, H, \text{Min})$ recording (day, month, year, hour, minute). Two encoders into $\mathcal{H}$ are defined:

$$(A) \text{ String-based: } \mathbf{T}_t = \xi_{\text{hv}}(\psi(\text{str}(F_t))), \quad (3.15)$$

$$(B) \text{ Compositional: } \mathbf{T}_t = \triangle(\rho^{D_{M,Y}} \mathbf{e}_{\text{day}}, \rho^M \mathbf{e}_{\text{month}}, \rho^H \mathbf{e}_{\text{hour}}, \rho^{\text{Min}} \mathbf{e}_{\text{min}}), \quad (3.16)$$

where $\mathbf{e}_{\text{day}}, \mathbf{e}_{\text{month}}, \mathbf{e}_{\text{hour}}, \mathbf{e}_{\text{min}} \in \mathcal{H}$ are pre-sampled basis hypervectors. In either case the field-bound vector is $|F_t\rangle = \mathbf{F}_t \otimes \mathbf{T}_t$.

Why approach (B) preserves temporal order

Theorem 5.5 — Temporal locality of encoder B

Theorem 3.21. *Let $\mathbf{T}_t$ and $\mathbf{T}'_t$ be encoded by approach (B) for two timestamps differing only in the hour field by ΔH . Then*

$$\mathbb{E}[\text{sim}(\mathbf{T}_t, \mathbf{T}'_t)] = \frac{3}{4} \mathbb{E}[\text{sim}(\mathbf{e}_{\text{hour}}, \rho^{\Delta H} \mathbf{e}_{\text{hour}})] + \frac{1}{4} + O(1/\sqrt{N}), \quad (3.17)$$

where $N = 4$ is the number of bundled fields. In particular the similarity decays monotonically with $|\Delta H|$ until $|\Delta H| \lesssim D$, after which it stabilises at the random level of $O(1/\sqrt{D})$.

Proof. Decompose the inner product $\langle \mathbf{T}_t, \mathbf{T}'_t \rangle$ by linearity of the dot product over the four bundled components. For every field except hour, the permutation indices match, so the contribution to the inner product is $\langle \rho^k \mathbf{e}, \rho^k \mathbf{e} \rangle = D$; summed across the three non-hour fields this contributes $3D$ to the inner product.

The hour terms contribute $\langle \rho^H \mathbf{e}_{\text{hour}}, \rho^{H+\Delta H} \mathbf{e}_{\text{hour}} \rangle = D \cdot \mathbb{E}[\text{sim}(\mathbf{e}_{\text{hour}}, \rho^{\Delta H} \mathbf{e}_{\text{hour}})]$ in expectation.

The bundling sign correction is $O(1/\sqrt{N})$ by Theorem 3.8. Dividing through by D and absorbing constants gives (3.17).

For the monotonicity: by Theorem 3.11(iii), as $|\Delta H| \rightarrow D$ (modulo D) the term $\text{sim}(\mathbf{e}_{\text{hour}}, \rho^{\Delta H} \mathbf{e}_{\text{hour}})$ tends to zero, and the overall similarity decays correspondingly. $\square$

Choice in the prototype

We adopt encoder (A) for the consensus and verification proofs (Sections 3.4 to 3.6) because it yields exact quasi-independence between distinct timestamps and is therefore analytically clean. Encoder (B) is retained as an optional layer that preserves temporal locality for downstream analytics in the spirit of [18].

3.3.3 Block-Hash Hypervector Encoding

Why a chain field is still kept

Although the architecture is *chainless* at the topological level (blocks are accepted by geometric similarity, not by pointer references), we retain a cryptographic hash $h_{B,n}$ that includes $h_{B,n-1}$ as input. This provides a fallback collision-resistant imprint of the full history, in the spirit of Antonopoulos [3] and the Bitcoin block header. Critically, this cryptographic chain is *redundant* for verification (which uses sim on $\mathbf{B}_n$ and $\mathbf{C}_n$); it is retained purely as defence-in-depth.

Construction

Definition 5.6 — Block-hash hypervector

Definition 3.22. For block n with identifier BlockID_n , timestamp t_n , previous-hash $h_{B,n-1}$, transaction bundle $\mathbf{Tx}_h$, and transaction count k , set

$$h_{B,n} = \psi(\text{BlockID}_n \parallel t_n \parallel h_{B,n-1} \parallel \psi(\mathbf{Tx}_h) \parallel k), \quad (3.18)$$

$$\mathbf{V}_n^h = \xi_{\text{hv}}(h_{B,n}), \quad |F_h\rangle = \mathbf{F}_h \otimes \mathbf{V}_n^h. \quad (3.19)$$

History sensitivity

Theorem 5.7 — History sensitivity

Theorem 3.23. Under the random-oracle model for ψ , modifying any ancestor block Bl_m with $m \leq n$ yields a perturbed sequence with the property that, with probability at least $1 - 2^{-256}$, $h'_{B,n} \neq h_{B,n}$ and consequently $\mathbf{V}_n^{h'}$ is quasi-orthogonal to $\mathbf{V}_n^h$: $\mathbb{E}[\text{sim}(\mathbf{V}_n^h, \mathbf{V}_n^{h'})] = 0$, $\text{Var} = 1/D$.

Proof. By construction, $h_{B,n}$ depends on $h_{B,n-1}$, which by induction depends on every prior $\text{BlockID}_m, t_m, h_{B,m-1}$. Modifying any of these inputs changes the input to ψ . Under the random-oracle assumption, the output of ψ on the new input is uniformly distributed and equals the original output with probability at most 2^{-256} . Conditional on $h'_{B,n} \neq h_{B,n}$, Theorem 3.15(P3) applies with $s = h_{B,n}$ and $s' = h'_{B,n}$, yielding the stated mean and variance. $\square$

Corollary 3.24 (Tamper-evident history without pointers). *A single bit-flip anywhere in the historical block sequence cascades, with overwhelming probability, into a hypervector $\mathbf{V}_n^{h'}$ that is geometrically indistinguishable from a freshly sampled random vector. The chain property is therefore encoded in $\mathcal{H}$ rather than in an explicit data structure.*

3.3.4 Chapter Summary

We have constructed three encoders—TXROOT, timestamp, and block-hash—that together produce field-bound hypervectors $|F_{tx}\rangle$, $|F_t\rangle$, $|F_h\rangle$ ready to be assembled into a block hypervector $\mathbf{B}_n$. The two quantitative theorems of the chapter, Theorem 3.19 (continuous tamper degradation of TXROOT) and Theorem 3.23 (history sensitivity of the block-hash hypervector), are the technical building blocks on which the verification predicate of Section 3.5 is built.

3.4 Block Hypervector, Context Vector, and Pairwise Link

This chapter assembles the field-bound hypervectors of Section 3.3 into a single block hypervector $\mathbf{B}_n \in \mathcal{H}$ (Section 3.4.1), defines the recursive *context vector* $\mathbf{C}_n$ that encodes the chain semantics geometrically (Section 3.4.2), and introduces the directed *pairwise geometric link* $L_{A \rightarrow B}$ between two blocks (Section 3.4.3). Each construction is followed by a quantitative theorem characterising its retrievability or tamper sensitivity.

3.4.1 The Block Hypervector $\mathbf{B}_n$

Definition 6.1 — Block hypervector

Definition 3.25. The *block hypervector* at slot n is the bundle

$$\mathbf{B}_n = \Delta(|F_h\rangle, |F_{tx}\rangle, |F_t\rangle, |F_r\rangle) \in \mathcal{H}, \quad (3.20)$$

where $|F_r\rangle$ is an optional extensible field-bound vector (e.g. proposer identity, slot number, gas usage, etc.). The number of bundled fields is denoted $N_f \geq 3$ and is fixed for the duration of the protocol.

Retrieval of a bound field

Theorem 6.2 — Field extraction by unbinding

Theorem 3.26. For any of the field-bound vectors $|F\rangle = \mathbf{F} \otimes \mathbf{V}$ with key $\mathbf{F} \in \{\mathbf{F}_h, \mathbf{F}_{tx}, \mathbf{F}_t, \mathbf{F}_r\}$,

$$\widehat{|F\rangle} := \mathbf{B}_n \otimes \mathbf{F} = \mathbf{V} + (\text{noise hypervector orthogonal to } \mathbf{V}), \quad (3.21)$$

and $\mathbb{E}[\text{sim}(\widehat{|F\rangle}, \mathbf{V})] = \Theta(1/\sqrt{N_f})$. For $N_f = 4$ and $D \geq 10^4$ the noise concentrates around its expectation and the query is reliable.

Proof. The binding operation $\otimes$ distributes over the (sign-corrected) bundle in expectation [5, 21]; this distributivity is the binding-bundling commutativity property of VSAs. Hence

$$\mathbf{B}_n \otimes \mathbf{F} = \Delta(\mathbf{F} \otimes |F_h\rangle, \dots, \mathbf{F} \otimes |F_r\rangle).$$

For the matching field, self-inverse of $\otimes$ (Theorem 3.5(v)) gives $\mathbf{F} \otimes (\mathbf{F} \otimes \mathbf{V}) = \mathbf{V}$. For each of the other $N_f - 1$ fields, $\mathbf{F} \otimes (\mathbf{F}' \otimes \mathbf{V}')$ is a hypervector quasi-orthogonal to $\mathbf{V}$ by Theorem 3.5(vi). The bundle of one signal and $N_f - 1$ orthogonal noises has expected cosine $\mathbb{E}\text{sim} = \Theta(1/\sqrt{N_f})$ by Theorem 3.8, which for $N_f = 4$ evaluates to $\sqrt{2/(4\pi)} \approx 0.40$ —well above any sensible threshold φ in $(0.1, 0.3)$. $\square$

Operational meaning

Theorem 3.26 states that the block hypervector $\mathbf{B}_n$ is, in a precise sense, a *key-value store*: each field can be retrieved by binding $\mathbf{B}_n$ with its key. This is the algebraic mechanism by which a single D -dimensional vector encodes all four block fields simultaneously, while remaining quasi-orthogonal to any unrelated block.

Worked Numerical Example 6.1 — Empirical field extraction

Fix $N_f = 4$ fields, $D = 10\,000$, and sample independent random keys $\mathbf{F}_1, \dots, \mathbf{F}_4$ and values $\mathbf{V}_1, \dots, \mathbf{V}_4$ uniformly from $\mathcal{H}$. Build the bundled-binding $\mathbf{B} = \text{sign}(\sum_{i=1}^4 \mathbf{F}_i \otimes \mathbf{V}_i)$.

Step 1 (theoretical predictions). Noise per unbinding has standard deviation $\sigma_{\text{noise}} = \sqrt{(N_f - 1)/D} = \sqrt{3/10\,000} \approx 0.0173$. The expected target similarity is $\sqrt{2/(N_f\pi)} = \sqrt{2/(4\pi)} \approx 0.399$.

Step 2 (Monte-Carlo result). Unbinding the bundle with the correct key $\mathbf{F}_1$ yields $\text{sim}(\mathbf{B} \otimes \mathbf{F}_1, \mathbf{V}_1) = 0.378$, while unbinding with an unrelated value vector gives $\text{sim}(\mathbf{B} \otimes \mathbf{F}_1, \mathbf{V}_2) = 0.004$, essentially the predicted Rademacher noise floor ± 0.01 .

Step 3 (false-retrieval bound). For any threshold $\varphi \in (0.1, 0.3)$, by Hoeffding's inequality applied to the noise component, $\mathbb{P}(|\text{noise similarity}| \geq \varphi) \leq \exp(-D\varphi^2/2(N_f - 1))$. For $D = 10\,000$, $N_f = 4$, $\varphi = 0.3$: $\exp(-10\,000 \cdot 0.09/6) = \exp(-150) \approx 7.18 \times 10^{-66}$. A gap of $0.378 - 0.004 = 0.374$ between true and false key is therefore statistically overwhelming.

Step 4 (scaling). Increasing the number of fields to $N_f = 16$ raises noise std to $\sqrt{15/D} \approx 0.0387$ and the false-retrieval bound becomes $\exp(-30) \approx 9.36 \times 10^{-14}$ at $\varphi = 0.3$, still negligible. This confirms the block hypervector remains a reliable key-value store as the number of stored fields grows.

3.4.2 The Context Vector $\mathbf{C}_n$

Definition 6.3 — Context vector

Definition 3.27. Let $\mathbf{C}_0 := \mathbf{v}_0$ be a *genesis hypervector* sampled uniformly from $\mathcal{H}$. Define recursively

$$\mathbf{C}_n = \Delta(\mathbf{C}_{n-1}, \mathbf{B}_n), \quad n = 1, 2, 3, \dots \quad (3.22)$$

Properties of the context vector

Theorem 6.4 — Properties of $\mathbf{C}_n$

Theorem 3.28. For every $n \geq 1$ the context vector $\mathbf{C}_n$ satisfies:

- (i) Geometric average. $\mathbf{C}_n \approx \text{sign}\left(\sum_{i=0}^n \mathbf{B}_i\right)$ in the sense that, up to ties of probability $O(2^{-n})$ per coordinate, $\mathbf{C}_n$ is the sign of an integer linear combination of $\mathbf{B}_0, \dots, \mathbf{B}_n$ with positive weights.
- (ii) Memory decay. For every fixed $k \leq n$, $\mathbb{E}[\text{sim}(\mathbf{C}_n, \mathbf{B}_{n-k})] = \Theta(1/\sqrt{n+1})$.
- (iii) Retroactive sensitivity. If $\mathbf{B}'_m \neq \mathbf{B}_m$ for some $m \leq n$, then with probability at least $1 - 2e^{-D\varepsilon^2/2}$,

$$\text{sim}(\mathbf{C}'_n, \mathbf{C}_n) \leq 1 - \frac{2}{n+1} + \varepsilon.$$

Proof. Proof of (i). Iterating the recursion (3.22) gives

$$\mathbf{C}_n = \text{sign}\left(\sum_{i=0}^n \alpha_i \mathbf{B}_i\right),$$

where the coefficients α_i are positive integers arising from the binary-tree structure of the nested sign operators. In Appendix B we exhibit the recursion $\alpha_i = 2^{n-i}$ for $i \geq 1$ and $\alpha_0 = 2^n$, modulo ties. Up to coordinate-wise ties of probability $O(2^{-n})$, the weights α_i are equal in distribution under the random-oracle model.

Proof of (ii). The hypervectors $\mathbf{B}_0, \dots, \mathbf{B}_n$ are i.i.d. uniform on $\mathcal{H}$ by Theorem 3.15(P3) applied to their distinct block-hash seeds. Apply Theorem 3.8 with $N = n+1$ to obtain the $\Theta(1/\sqrt{n+1})$ rate.

Proof of (iii). Tampering with block m replaces $\mathbf{B}_m$ by $\mathbf{B}'_m$, which is independent of $\mathbf{B}_m$ by the random-oracle property (Theorem 3.23). Coordinate-wise, the probability that $\text{sign}(\sum_i \alpha_i \mathbf{B}_{i,j}) = \text{sign}(\sum_i \alpha_i \mathbf{B}'_{i,j})$ is $1 - 1/(n+1) + o(1)$, because the perturbation alters $\alpha_m \approx 2^{n-m}$ of the total weight $\sum_i \alpha_i \approx 2^{n+1}$. Averaging across D coordinates and applying Hoeffding's inequality gives the stated tail bound. $\square$

Operational meaning

$\mathbf{C}_n$ is the *historical fingerprint* of the chain. Property (ii) says that no single block dominates the fingerprint (the influence of every block decays as $1/\sqrt{n+1}$), so the context is a robust summary. Property (iii) says that tampering with any past block leaves a detectable geometric signature in $\mathbf{C}_n$: the similarity between the original and tampered contexts drops by at least $2/(n+1)$, which is exactly the signal exploited by the verification predicate of Section 3.5.

Worked Numerical Example 6.2 — Context memory decay and tamper signal

Consider the weighted-context variant $\mathbf{C}_n = \sum_{j=0}^W \alpha^j \mathbf{B}_{n-j}$ with decay factor $\alpha = 0.7$ and window $W = 5$ (see also Appendix B):

lag j	weight α^j	% of total weight
0	1.0000	34.81%
1	0.7000	24.37%
2	0.4900	17.06%
3	0.3430	11.94%
4	0.2401	8.36%
5	0.1681	5.85%
sum	2.873	100.00%

Tamper detection at depth j . If an attacker rewrites block $\mathbf{B}_{n-j}$, the expected drop in $\text{sim}(\mathbf{C}_n, \mathbf{C}'_n)$ is approximately twice the relative weight: $\Delta \text{sim}(j) \approx 2\alpha^j / \sum_{i=0}^W \alpha^i$. Thus tampering with the latest block ($j = 0$) drops similarity by ≈ 0.696 ; with $j = 3$ by 0.239; with $j = 5$ by only 0.117. Older tampering is harder to detect via $\mathbf{C}_n$ alone, but combined with the per-block predicate $\mathcal{V}(\mathbf{B}_n)$ this two-level architecture (Section 3.5) still enforces $\Theta(D)$ -level soundness across all depths.

Geometric average variant. For the unweighted geometric average $\mathbf{C}_n = \text{sign}(\sum_{i \leq n} \mathbf{B}_i)$, by Theorem 3.8 with $N = n+1$, the expected similarity of a single block to the context decays as $\sqrt{2/(\pi(n+1))}$:
 $n = 1$: 0.564, $n = 10$: 0.241, $n = 100$: 0.080, $n = 10\,000$: 0.008. No single block

dominates the long-run fingerprint, confirming Property (ii).

3.4.3 Pairwise Geometric Link $L_{A \rightarrow B}$

Connection with HRR

Plate’s HRR [17, 6] represents a directed association $A \rightarrow B$ by the bound vector $A \circledast \rho B$, where $\circledast$ denotes circular convolution. We adapt this construction to BSC by replacing circular convolution with elementwise binding, yielding the directed pairwise link below.

Definition 6.5 — Directed pairwise link

Definition 3.29. For blocks A, B with hypervectors $\mathbf{B}_A, \mathbf{B}_B \in \mathcal{H}$, the *directed link from A to B* is

$$L_{A \rightarrow B} := \mathbf{B}_A \otimes \rho(\mathbf{B}_B) \in \mathcal{H}. \quad (3.23)$$

Theorem 6.6 — Properties of $L_{A \rightarrow B}$

Theorem 3.30. *The directed link satisfies:*

- (i) Asymmetry. $L_{A \rightarrow B} \neq L_{B \rightarrow A}$ in general, because $\rho \mathbf{B}_B \neq \mathbf{B}_B$.
- (ii) Retrieval. $\rho^{-1}(\mathbf{B}_A \otimes L_{A \rightarrow B}) = \mathbf{B}_B + (\text{quasi-orthogonal noise})$.
- (iii) Path composition. $L_{\text{path}} := \triangle(L_{A \rightarrow B}, L_{B \rightarrow C})$ is a hypervector that contains $\mathbf{B}_B$ -noise but preserves the binding witnesses $\mathbf{B}_A \otimes \rho \mathbf{B}_B$ and $\mathbf{B}_B \otimes \rho \mathbf{B}_C$ in the sense that each can be recovered by binding with the appropriate $\mathbf{B}_A$ or $\mathbf{B}_B$.

Proof. Proof of (i). We compute $\text{sim}(L_{A \rightarrow B}, L_{B \rightarrow A}) = \text{sim}(\mathbf{B}_A \otimes \rho \mathbf{B}_B, \mathbf{B}_B \otimes \rho \mathbf{B}_A)$. By Theorem 3.11(iii), $\rho \mathbf{B}_A$ is uncorrelated with $\mathbf{B}_A$ in expectation, hence the right-hand side has mean 0 and variance $1/D$. Thus the two links are quasi-orthogonal and in particular distinct.

Proof of (ii). Using self-inverse of $\otimes$ (Theorem 3.5(v)), $\mathbf{B}_A \otimes L_{A \rightarrow B} = \mathbf{B}_A \otimes (\mathbf{B}_A \otimes \rho \mathbf{B}_B) = \rho \mathbf{B}_B$. Applying ρ^{-1} recovers $\mathbf{B}_B$ exactly.

Proof of (iii). Bundling commutes pointwise with binding only up to the sign-correction of Theorem 3.8. The retrieval bound of (ii) is therefore degraded by the bundling similarity drop of Theorem 3.8 with $N = 2$. The expected retrieval similarity is $\sqrt{2/(2\pi)} = \sqrt{1/\pi} \approx 0.56$, well above the quasi-orthogonality floor. $\square$

When is $L_{A \rightarrow B}$ useful?

The pairwise link is not strictly required by the verification predicate of Section 3.5, which uses $\mathbf{B}_n$ and $\mathbf{C}_n$ alone. Its purpose is structural: it permits *ad hoc* queries of the form “did block B ever follow block A ?”, answered by computing $\text{sim}(L_{A \rightarrow B}, \mathbf{B}_A \otimes \rho \mathbf{B}_B)$ on locally recomputed hypervectors. Such queries are useful for forensic analysis and for cross-shard interoperability protocols beyond the scope of this work.

3.4.4 Chapter Summary

The block hypervector $\mathbf{B}_n$ is a D -dimensional key-value store of the four block fields, each retrievable by binding (Theorem 3.26). The context vector $\mathbf{C}_n$ is a recursively bundled summary of the historical chain, with no single block dominating, and a quantifiable sensitivity to any retroactive tampering (Theorem 3.28). The directed pairwise link $L_{A \rightarrow B}$ supplements these with an asymmetric pointer that preserves the BSC algebra. These three primitives feed directly into the similarity-based verification predicate of Section 3.5.

3.5 Similarity-Based Verification

This chapter is the analytical heart of this work: we develop a *two-level* verification predicate $\mathcal{V}$ that accepts or rejects a block $\mathbf{B}_n$ on the basis of two cosine similarities—block-level S_B and context-level S_C . The predicate is proved to have type-I and type-II error probabilities that decay exponentially in the hyperdimension D (Theorem 3.32), and the resulting consensus rule is shown to be statistically sound for every fixed tamper magnitude $m \geq 1$ as $D \rightarrow \infty$ (Theorem 3.34).

3.5.1 Two-Level Verification Scores

Definition 7.1 — Block- and context-level scores

Definition 3.31. Given a locally re-derived block hypervector $\mathbf{B}_n^L$ (computed from a validator's own state) and a locally re-derived context vector $\mathbf{C}_n^L$, define

$$S_B := \text{sim}(\mathbf{B}_n^L, \mathbf{B}_n) = \frac{1}{D} \sum_{j=1}^D B_{n,j}^L B_{n,j}, \quad S_C := \text{sim}(\mathbf{C}_n^L, \mathbf{C}_n'). \quad (3.24)$$

The score S_B measures whether the proposer's block matches the validator's local re-derivation of the block contents. The score S_C measures whether the proposer's claimed new context vector $\mathbf{C}_n' = \Delta(\mathbf{C}_{n-1}, \mathbf{B}_n)$ is consistent with the validator's local context $\mathbf{C}_n^L$.

3.5.2 Decision-Theoretic Error Bounds

We model two hypothesis classes:

- H_0 — *block is genuine*. The proposer constructed $\mathbf{B}_n$ honestly. Then $S_B \approx 1 - \delta$ for some small $\delta \geq 0$ that accounts for unavoidable rounding/sign-tie errors; in practice $\delta = O(D^{-1/2})$.
- H_1 — *block is tampered*. The proposer altered at least $m \geq 1$ of the N_f field-bound vectors. By the tamper-sensitivity argument of Theorem 3.19 combined with the distributivity of binding over bundling (Theorem 3.26), $\mathbb{E}[S_B] \leq 1 - 2m/N_f + O(N_f^{-1/2})$.

Theorem 7.2 — ROC bounds

Theorem 3.32. Fix any threshold $\varphi_1 \in (0, 1)$. Then the type-I and type-II error probabil-

ities of the predicate $S_B > \varphi_1$ satisfy the Hoeffding bounds

$$\mathbb{P}_{H_0}(S_B < \varphi_1) \leq \exp(-\tfrac{1}{2}D(1 - \delta - \varphi_1)^2), \quad (3.25)$$

$$\mathbb{P}_{H_1}(S_B > \varphi_1) \leq \exp(-\tfrac{1}{2}D(2m/N_f - 1 + \varphi_1)^2). \quad (3.26)$$

The analogous bounds hold with S_B replaced by S_C .

Proof. S_B is the mean of D random variables in $[-1, +1]$ given by the per-coordinate products $B_{n,j}^L B_{n,j}$.

Under H_0 the expectation of each such product is $1 - \delta$, and Hoeffding's one-sided inequality applied to the deviation $S_B - \mathbb{E}_{H_0}[S_B] = S_B - (1 - \delta)$ gives $\mathbb{P}_{H_0}(S_B < \varphi_1) = \mathbb{P}_{H_0}(S_B - (1 - \delta) < \varphi_1 - (1 - \delta)) \leq e^{-\frac{1}{2}D(1 - \delta - \varphi_1)^2}$, which is (3.25).

Under H_1 , $\mathbb{E}_{H_1}[S_B] \leq 1 - 2m/N_f + o(1)$. By a symmetric application of Hoeffding's inequality,

$$\mathbb{P}_{H_1}(S_B > \varphi_1) \leq e^{-\frac{1}{2}D(\varphi_1 - (1 - 2m/N_f))^2} = e^{-\frac{1}{2}D(2m/N_f - 1 + \varphi_1)^2},$$

which is (3.26). The same argument applies verbatim to S_C , where the analogous expectations follow from Theorem 3.28(iii). $\square$

Optimal threshold and equal-error rate

A standard receiver-operating-characteristic (ROC) trade-off prescribes that we choose φ_1 at the *equal-error rate* (EER): the value at which type-I and type-II error probabilities coincide. Equating the exponents in (3.25) and (3.26) gives

$$(1 - \delta - \varphi_1)^2 = (2m/N_f - 1 + \varphi_1)^2,$$

which (taking the positive root) yields the closed-form solution

$$\varphi_1^* = \frac{(1 - \delta) + (1 - 2m/N_f)}{2} = 1 - \frac{\delta + 2m/N_f}{2}. \quad (3.27)$$

At $\varphi_1 = \varphi_1^*$ both error probabilities are bounded by $\exp(-\frac{1}{2}D \cdot (\frac{1 - \delta - 2m/N_f}{2})^2) = e^{-cD}$, with a constant $c = c(m, N_f, \delta) > 0$ whenever $1 - \delta - 2m/N_f > 0$, i.e. whenever the genuine and tampered expectations are separable.

Worked Numerical Example 7.1 — False-accept rate table

Consider the simplified impostor model in which S_B for a tampered block is approximately $\mathcal{N}(0, 1/D)$ (Hoeffding regime). Then the false-accept rate at threshold φ is bounded by $\text{FAR}(\varphi) \leq \exp(-D\varphi^2/2)$.

$D \setminus \varphi$	0.05	0.10	0.15	0.20
1 000	2.87×10^{-1}	6.74×10^{-3}	1.78×10^{-5}	2.06×10^{-9}
10 000	3.73×10^{-6}	1.93×10^{-22}	5.95×10^{-49}	1.38×10^{-87}
100 000	5.17×10^{-55}	7.13×10^{-218}	≈ 0	≈ 0

The prototype operating point $(D, \varphi) = (10\,000, 0.10)$ achieves $\text{FAR} \leq 1.93 \times 10^{-22}$, i.e. less than one in 10^{21} tampered blocks would slip past the per-block test. Over the entire lifetime of Bitcoin to date ($\approx 10^6$ blocks), even a Sybil adversary submitting one impostor per block would expect 10^{-16} false accepts in total.

Worked Numerical Example 7.2 — ROC and EER calibration

Fix the operating point $(m, N_f, \delta) = (1, 4, 0.05)$: a single field altered out of four, and a 5% block-level genuine-similarity slack. The EER threshold from (3.27) is $\varphi_1^* = 1 - \frac{0.05+2/4}{2} = 1 - 0.275 = 0.725$. The EER itself is bounded by $\exp\left(-\frac{1}{2}D \cdot \left(\frac{1-0.05-0.5}{2}\right)^2\right) = \exp(-D \cdot 0.0506)$.

D	EER bound at $\varphi^* = 0.725$	Interpretation
1 000	$\approx 6.32 \times 10^{-23}$	cryptographically negligible
10 000	$\approx \exp(-506) \approx 10^{-220}$	overkill
100 000	$\approx \exp(-5\,063)$	overkill

Conclusion. For the prototype parameters $(D, \varphi_1^*) = (10\,000, 0.725)$, both type-I (false reject) and type-II (false accept) errors are bounded by 10^{-220} , far below any conceivable adversarial budget. The two-level verification predicate is therefore statistically sound by an extraordinary margin.

3.5.3 The Consensus-Level Acceptance Predicate

Definition 7.3 — Block acceptance predicate

Definition 3.33. The (two-level) *block acceptance predicate* is

$$\mathcal{V}(\mathbf{B}_n) = \mathbf{1}[S_B > \varphi_1 \wedge S_C > \varphi_2], \quad \varphi_1, \varphi_2 \in (0, 1). \quad (3.28)$$

Corollary 7.4 — Geometric soundness

Corollary 3.34. Choose φ_1, φ_2 at the equal-error rate of the respective ROC curves. Then $\mathcal{V}$ accepts genuine blocks and rejects m -tampered blocks with error probability at most $\exp(-cD)$ for a positive constant $c = c(m, N_f, \delta)$. The system is therefore statistically sound for every fixed $m \geq 1$ as $D \rightarrow \infty$.

Proof. Under H_0 a genuine block produces $S_B > \varphi_1^*$ and $S_C > \varphi_2^*$ with probability at least $1 - 2e^{-cD}$ by Theorem 3.32. Under H_1 a tampered block triggers at least one of $S_B < \varphi_1^*$ or $S_C < \varphi_2^*$ with the same exponential bound. The conjunction in (3.28) therefore achieves the stated error rate by a union bound. $\square$

3.5.4 Numerical Calibration

Take the prototype parameters $D = 10\,000$, $N_f = 4$, and $\delta = 0$ (no rounding error). For tamper magnitudes $m \in \{1, 2, 3\}$ we obtain

Table 3.1: Numerical predictions for the two-level verification predicate at the equal-error rate (3.27). Error probabilities are upper bounds from Theorem 3.32.

m	$\mathbb{E}_{H_1}[S_B]$	φ_1^*	Error bound $\leq$
1	0.50	0.75	$e^{-312.5} \approx 10^{-136}$
2	0.00	0.50	$e^{-1250} \approx 10^{-543}$
3	-0.50	0.25	$e^{-2813} \approx 10^{-1222}$

Table 3.1 confirms that even the weakest tamper case (a single altered field out of four) is rejected with a cosmologically small failure probability at $D = 10\,000$. In practice, the *network* reliability dwarfs the *geometric* reliability of $\mathcal{V}$, which is exactly what one wants in a verification primitive.

3.5.5 Chapter Summary

The two-level predicate $\mathcal{V}$ accepts a block if and only if both the block-level cosine similarity S_B and the context-level cosine similarity S_C exceed thresholds φ_1, φ_2 chosen at the equal-error rate of the respective ROC curves. Under standard SHA-256 and BSC assumptions, the type-I and type-II error probabilities are bounded by $\exp(-cD)$ for a positive c depending on the tamper magnitude m and the number of fields N_f . The predicate is the admissibility criterion of the PoS-HGV protocol presented in Section 3.6.

3.6 PoS Consensus with HGV Verification

This chapter integrates the geometric verification predicate $\mathcal{V}$ of Section 3.5 with a standard Tendermint-style proof-of-stake (PoS) Byzantine fault-tolerant (BFT) protocol. We prove that the resulting *PoS-HGV* protocol is safe (Theorem 3.36) and live (Theorem 3.37) under the standard partial-synchrony and $< 1/3$ -Byzantine-stake assumptions, modulo a geometric error term that decays exponentially in D .

3.6.1 Context and Design Choice

Tendermint Core [10, 27] and related PoS BFT protocols [16, 15] achieve safety provided fewer than one-third of the staked validators are Byzantine. We adopt their voting machinery essentially verbatim with a *single* modification: a validator’s vote is ADMISSIBLE only if its local geometric verification $\mathcal{V}(\mathbf{B}_n) = 1$. Everything else—the leader-selection rule weighted by stake, the two-thirds majority threshold, the gossip protocol—is inherited.

3.6.2 Protocol Specification

Definition 8.1 — PoSHGV slot- n protocol

Definition 3.35. At slot n the protocol proceeds in six steps.

1. *Leader selection.* A leader $\mathcal{L}_n$ is sampled with probability proportional to stake from the set of active validators.

2. *Block creation.* The leader builds $\mathbf{B}_n$ according to Theorem 3.25 and proposes the new context $\mathbf{C}'_n := \Delta(\mathbf{C}_{n-1}, \mathbf{B}_n)$.
3. *Broadcast.* The pair $(\mathbf{B}_n, \mathbf{C}'_n)$ is gossiped to all validators.
4. *Local validation.* Each validator recomputes its own $\mathbf{B}_n^L$ and $\mathbf{C}_n^L$ from its local state and measures the cosine similarities S_B, S_C of Theorem 3.31.
5. *Voting.* A validator votes ACCEPT if and only if $\mathcal{V}(\mathbf{B}_n) = 1$ (i.e. both similarity thresholds pass) and every included transaction is valid under the application semantics.
6. *Commit.* If at least 2/3 of the staked weight votes ACCEPT, the block is committed: $\mathbf{B}_n$ is appended to the ledger, $\mathbf{C}_n := \mathbf{C}'_n$ is adopted, and $\mathcal{L}_n$ is rewarded.

Algorithm 1 PoS-HGV validator routine at slot n .

Require: Local state $\mathcal{S}_n^L$; thresholds φ_1, φ_2 ; received block $\mathbf{B}_n$ and proposed context $\mathbf{C}'_n$.

- 1: $\mathbf{B}_n^L \leftarrow \text{LOCALBUILD BLOCK}(\mathcal{S}_n^L, n)$
 - 2: $\mathbf{C}_n^L \leftarrow \Delta(\mathbf{C}_{n-1}, \mathbf{B}_n^L)$
 - 3: $S_B \leftarrow \text{sim}(\mathbf{B}_n^L, \mathbf{B}_n)$
 - 4: $S_C \leftarrow \text{sim}(\mathbf{C}_n^L, \mathbf{C}'_n)$
 - 5: **if** $S_B > \varphi_1$ **and** $S_C > \varphi_2$ **and** $\text{ALLTXVALID}(\mathbf{B}_n)$ **then**
 - 6: VOTE $\leftarrow$ accept
 - 7: **else**
 - 8: VOTE $\leftarrow$ reject
 - 9: **end if**
 - 10: **return** VOTE
-

3.6.3 Safety

Theorem 8.2 — Safety

Theorem 3.36. *Assume:*

- (a) *the network is partially synchronous [11];*
- (b) *less than 1/3 of the total staked weight is held by Byzantine validators;*
- (c) *the hyperdimension D is large enough that the per-block ROC error of $\mathcal{V}$ satisfies $\eta := \mathbb{P}_{H_0}(S_B < \varphi_1) + \mathbb{P}_{H_1}(S_B > \varphi_1) < 1/3$.*

Then the PoS-HGV protocol of Theorem 3.35 is safe: no two honest validators commit different blocks at the same slot n , except on an event of probability $\leq 2e^{-cD}$ for a positive constant c .

Proof. Standard Tendermint safety [10, 27] guarantees that, under assumptions (a) and (b), no two honest validators commit different blocks at slot n provided that the validity predicate

is identical at every honest validator. The only deviation from textbook Tendermint is the addition of the geometric verification step $\mathcal{V}$ in line 5 of Theorem 3.35.

The remaining failure mode is that a malicious leader produces a tampered $\mathbf{B}_n$ that nevertheless passes $\mathcal{V}$ at $\geq 1/3$ honest validators (so that, combined with the Byzantine $< 1/3$, a fraudulent $\geq 2/3$ quorum is achieved). By Theorem 3.32 this event has probability at most $\binom{n_h}{n_h/3} \cdot \mathbb{P}_{H_1}(S_B > \varphi_1)^{n_h/3}$, where n_h is the number of honest validators. Each individual event has probability $\leq e^{-D(2/N_f - 1 + \varphi_1)^2/2}$, and the union bound across the $\binom{n_h}{n_h/3} \leq 2^{n_h}$ voting subsets gives the overall bound $2^{n_h} \cdot e^{-Dc'} \leq 2e^{-cD}$ for some positive $c = c'/2$ once D is large enough to dominate $n_h \cdot \log 2$. Combining this with the standard Tendermint safety argument yields the stated guarantee. $\square$

Worked Numerical Example 8.1 — Per-block error and dimension selection

The per-block verification error in PoS-HGV is $\varepsilon_v(D, \varphi) \leq 2 \exp(-\frac{1}{2}D\varphi^2)$. For BFT safety to absorb the geometric error, we need $\varepsilon_v < 1/3$.

D	$\varphi = 0.05$	$\varphi = 0.10$	$\varphi = 0.20$	$< 1/3?$
1 000	0.573	1.35×10^{-2}	4.12×10^{-9}	$\varphi \geq 0.10$
10 000	7.45×10^{-6}	3.86×10^{-22}	2.77×10^{-87}	all $\varphi \geq 0.05$
100 000	1.03×10^{-54}	2.85×10^{-217}	≈ 0	all

Minimum D for cryptographic-grade safety. To achieve $\varepsilon_v < 10^{-9}$ at $\varphi = 0.10$ we solve $2 \exp(-D \cdot 0.005) < 10^{-9} \iff D > \frac{\ln(2 \cdot 10^9)}{0.005} \approx 4\,283$. Thus $D = 5\,000$ already exceeds the cryptographic threshold; the prototype choice $D = 10\,000$ provides over 13 orders of magnitude of additional margin.

Compounded safety over a chain of N blocks. By the union bound, the probability of any verification failure across N consecutive blocks at $(D, \varphi) = (10\,000, 0.10)$ is

$N \cdot \varepsilon_v \leq N \cdot 3.86 \times 10^{-22}$.	
N (blocks)	$\mathbb{P}(\text{any error}) \leq$
10^3	3.86×10^{-19}
10^6	3.86×10^{-16}
10^9	3.86×10^{-13}

Even across 10^9 blocks — roughly 1,900 years of Bitcoin-rate production — the union-bound failure probability remains under 4×10^{-13} . PoS-HGV at $D = 10\,000$ is therefore safe with overwhelming probability for any practical operational horizon.

3.6.4 Liveness

Theorem 8.3 — Liveness

Theorem 3.37. *Under the assumptions of Theorem 3.36, the PoS-HGV protocol of Theorem 3.35 is live: a new block is committed in expected $O(1)$ rounds.*

Proof. Tendermint liveness [10, 27] establishes $O(1)$ expected rounds to commit under partial synchrony and $< 1/3$ Byzantine stake. The only step added by PoS-HGV is the local com-

putation of $\mathcal{V}$ in line 5 of Theorem 3.35, which costs two D -length cosine similarities and one ALLTXVALID check. Each operation is constant-time per validator (linear in D and in the number of transactions). Hence the added latency is bounded uniformly in n , preserving the $O(1)$ round complexity. $\square$

3.6.5 Complexity Analysis

Corollary 8.4 — $O(1)$ verification complexity

Corollary 3.38. *Each validator performs at most two D -length cosine similarities per block, irrespective of the slot index n . The verification time is therefore $O(D) = O(1)$ in n , addressing the linear-history bottleneck of standard chain-based verification [3, 4].*

Proof. The two scores S_B and S_C each require a D -length dot product, totalling $2D$ scalar multiplications and $2D$ scalar additions. No additional dependency on n is introduced. The update $\mathbf{C}_n := \Delta(\mathbf{C}_{n-1}, \mathbf{B}_n)$ is itself an $O(D)$ operation. The local re-derivation $\mathbf{B}_n^L$ requires processing the k transactions of slot n , which is independent of n (it is the per-slot work that any PoS protocol must perform). $\square$

Remark 3.39 (Comparison with canonical verification). Bitcoin and Ethereum full nodes must re-verify each block in the chain to catch up to the tip. With $n \sim 10^7$ blocks, this is a $\Theta(n)$ workload measured in days. The chainless hyperdimensional ledger replaces this dependency with a single $O(D)$ cosine similarity. The trade-off is a higher per-block constant ($D = 10\,000$ scalar operations versus a single 256-bit hash), but the cross-over point is reached at $n \approx 10^5$ blocks: well below the size of any deployed chain.

3.6.6 Chapter Summary

The PoS-HGV protocol inherits Tendermint’s safety and liveness guarantees under the standard $< 1/3$ -Byzantine assumption and adds an $O(D)$ geometric verification step whose type-I/II errors decay exponentially in D . The per-block verification cost is independent of the slot index n , breaking the linear-history dependency of canonical chain-based ledgers. This concludes the technical core of this work; Section 3.7 examines the implications and limitations.

3.7 Discussion

This chapter steps back from the technical formalism of Sections 3.1 to 3.6 and discusses three broader themes: a quantitative comparison of the chainless architecture against Bitcoin, Ethereum, and Tendermint (Section 3.7.1); the security model and its boundaries (Section 3.7.2); and directions for future work (Section 3.7.3).

3.7.1 Comparison with Existing Architectures

Table 3.2 summarises the comparison along five dimensions: per-block verification cost, per-block storage, tamper signal, history dependency, and finality model. The proposed architecture

wins on the first four dimensions and is competitive on the fifth.

Table 3.2: Quantitative comparison of architectures. n is the slot index; k is the number of transactions per block; D is the hyperdimension.

	Bitcoin	Ethereum	Tendermint	PoS-HGV (this work)
Per-block verify	$O(\log k)$ inclusion, $\Theta(n)$ full re-verify	$O(\log k)$ Verkle, $\Theta(n)$ full	$O(\log k) + 2/3$ vote	$O(D)$, indep. of n
Per-block storage	~ 1 KB header + body	~ 1 KB header + body	~ 1 KB header + body	D bipolar bits = $D/8$ bytes
Tamper signal	Binary	Binary	Binary	Continuous: $1 - 2m/k$
History dep. on n	Linear	Linear	Linear (PBFT log)	None
Finality	Probabilistic ($\Theta(q^k)$)	Probabilistic (PoS) or final (Casper)	Deterministic	Deterministic

Storage. For $D = 10\,000$, a block hypervector occupies $D/8 = 1250$ bytes, comparable to a Bitcoin block header (80 bytes) plus its Merkle root (32 bytes) plus the included transactions. Crucially, the hypervector itself encodes *all* fields simultaneously (transactions, timestamp, hash, optional data), so it is closer to a full “block summary” than to a header alone.

Throughput. PoS-HGV inherits Tendermint’s throughput, which in practical deployments approaches 10^3 – 10^4 transactions per second under reasonable network conditions. The geometric verification step adds $O(D)$ floating-point operations per block, which is negligible against the network-bound cost of vote collection.

Energy. Like all PoS systems, PoS-HGV avoids the ~ 130 TWh/year energy footprint of Bitcoin [12]. The added geometric step is bandwidth- and CPU-bound and contributes no measurable energy overhead.

3.7.2 Security Analysis Boundaries

What this work proves

The safety theorem (Theorem 3.36) establishes that, under the random-oracle model for SHA-256 and the Tendermint partial synchrony / $< 1/3$ Byzantine stake assumptions, no two honest validators commit different blocks at the same slot except on an event of probability $\leq 2e^{-cD}$. For $D = 10\,000$ this is cosmologically small.

The tamper sensitivity theorem (Theorem 3.19) quantifies the cosine similarity degradation as $1 - 2m/k$ for m altered transactions out of k , with concentration error $O(e^{-D\epsilon^2/2})$. The retroactive sensitivity of the context vector (Theorem 3.28(iii)) gives the analogous statement at the chain-history level.

What this work does not prove

Computational vs. information-theoretic. All security claims here are *computational*: they reduce to the pre-image and collision resistance of SHA-256 and to the IND-CPA security

of the underlying stream PRNG. A quantum adversary running Grover’s algorithm reduces SHA-256 pre-image security from 2^{256} to 2^{128} , which is still comfortably above any reasonable attack budget. The hypervector algebra adds no quantum vulnerability beyond that of the underlying hash.

Network-level attacks. Eclipse attacks, BGP hijacking, and selfish-mining-style attacks against PoS leader selection are orthogonal to the geometric construction and are not analysed here. They are inherited from the Tendermint deployment context.

Sybil resistance. PoS-HGV relies on PoS as the Sybil-resistance mechanism. A naive chainless architecture without PoS would be trivially Sybil-able: an adversary could mint arbitrarily many genesis hypervectors and claim any history. This is the reason for retaining the standard PoS leader-selection rule of Theorem 3.35(1).

Side channels. The cosine similarity computation is data-dependent in the sense that its arithmetic intensity depends on the bit pattern of the hypervectors. Side-channel resistance is an implementation matter beyond the scope of the present analysis.

Adversary model summary

The adversary modelled here is:

- Polynomial-time (cannot invert SHA-256 better than brute force);
- Aware of all public parameters (the field keys $\mathbf{F}_h, \mathbf{F}_{tx}, \mathbf{F}_t, \mathbf{F}_r$, the genesis hypervector $\mathbf{C}_0$, the hyperdimension D);
- Able to corrupt fewer than $1/3$ of the staked weight;
- Subject to a partially synchronous network.

Under this model the safety and liveness theorems of Section 3.6 hold with the geometric error term decaying exponentially in D .

3.7.3 Limitations and Future Work

Empirical validation

The treatment is purely theoretical. A natural next step is to implement the architecture on a hardware testbed and measure the empirical error rates against the Hoeffding predictions of Table 3.1. We expect the empirical errors to be *lower* than the analytical upper bounds because Hoeffding is rarely tight, but the predicted exponential decay in D should be observable directly.

Hardware accelerators

HDC primitives are natively suited to hyperdimensional hardware [18, 7]. A FPGA or ASIC implementation of the verification predicate $\mathcal{V}$ should achieve sub-microsecond per-block latency

for $D = 10\,000$, making PoS–HGV competitive with state-of-the-art PoS systems even after accounting for the wider per-block payload.

Integration with Quantum Computing

The most ambitious direction for future work—and the principal follow-up to the present treatment—is the integration of the Hyperblock architecture with quantum computation. The bipolar hypervector space $\mathcal{H} = \{-1, +1\}^D$ is mathematically isomorphic to the computational basis of a D -qubit Hilbert space $(\mathbb{C}^2)^{\otimes D}$ under the natural embedding $+1 \mapsto |0\rangle$, $-1 \mapsto |1\rangle$. This correspondence suggests several concrete research avenues.

(i) Quantum HDC primitives. Binding ($\otimes$) becomes a tensor product of single-qubit Pauli- Z rotations; bundling (Δ) admits a coherent realisation via the quantum amplitude estimation primitive of Brassard et al. [28]. A quantum implementation of ξ_{hv} could prepare a superposition over hypervector candidates in $O(\log D)$ gate depth, opening the door to Grover-style acceleration of batch verification.

(ii) Quantum cosine-similarity estimation. The two-level verification predicate $\mathcal{V}$ reduces to two D -dimensional inner products. The swap test of Buhrman et al. [29] estimates $\langle \psi | \phi \rangle^2$ to precision ε in $O(1/\varepsilon^2)$ swap-test repetitions and a circuit depth of $O(\log D)$, providing a quadratic speedup over classical $O(D)$ similarity computation. For $D = 10\,000$, this reduces the per-block verification cost from $\sim 10^4$ classical multiplications to $\sim 10^2$ quantum swap tests at $\varepsilon = 10^{-2}$ precision—more than sufficient for the threshold $\varphi_1^* = 0.725$ derived in the EER calibration of Section 3.5.2.

(iii) Quantum random number generation (QRNG) for ξ_{hv} . The seeded map ξ_{hv} depends on a pseudo-random number generator stretched from the SHA-256 output (Theorem 3.13). Although computationally indistinguishable from uniform, this PRNG is a *deterministic* expansion of a finite seed and is therefore in principle backtrackable by an adversary who learns the seed. A quantum random number generator (QRNG) replaces this PRNG with a hardware source of *information-theoretically* unpredictable bits derived from a physical quantum process: vacuum-field fluctuations [30], photon path superposition, or shot-noise sampling in a balanced homodyne detector. National metrology institutes (e.g. NIST, PTB) and commercial providers (ID Quantique, Quantinuum, Quside) now certify QRNG output rates in the multi-Gb/s regime with min-entropy ≥ 0.99 bits/sample [31].

Integrating QRNG into ξ_{hv} has three concrete benefits:

1. *True randomness for hypervector coordinates.* Each of the $D = 10\,000$ bipolar entries of $\xi_{\text{hv}}(s)$ becomes the sign of a physically random bit, removing the residual PRNG-distinguishability gap noted in the remark following Theorem 3.13.
2. *Information-theoretic unpredictability of leader selection.* The PoS–HGV leader-selection step (Theorem 3.35) uses ξ_{hv} -derived randomness to break stake-weighted ties; replacing the PRNG with QRNG makes the leader sequence unpredictable even against an adversary with unbounded classical compute.

3. *Forward-secret seed material.* A QRNG-seeded ξ_{hv} admits per-block fresh randomness without relying on a long-lived master seed, eliminating the backtracking attack surface entirely.

The random-oracle proofs of Section 3.2 remain valid under a QRNG substitution; in fact they become *tight*, since the idealised uniform-PRNG assumption used throughout is now realised exactly rather than computationally.

(iv) **Post-quantum security of ξ_{hv} .** The security of ξ_{hv} rests on the random-oracle behaviour of SHA-256. Grover’s algorithm reduces SHA-256’s effective pre-image security from 2^{256} to 2^{128} quantum queries—still computationally infeasible for several decades but no longer *cosmologically* safe. A natural extension is to instantiate ξ_{hv} with a post-quantum hash family such as SPHINCS⁺ or a lattice-based PRF, preserving the random-oracle proofs of Section 3.2 unchanged.

(v) **Quantum consensus protocols.** The PoS-HGV protocol of Theorem 3.35 requires honest validators to broadcast vote messages. Replacing this with a quantum Byzantine agreement protocol (e.g. Ben-Or–Hassidim [32]) could give exponentially small error in the number of qubits per message, complementing the exponential geometric error in D . The combined system would enjoy a doubly-exponential safety margin, at the cost of requiring distributed entanglement between validators—a hardware capability that, while not presently practical, is on the roadmap of several quantum-network initiatives.

(vi) **Hyperblock as a quantum-classical bridge.** Finally, because $\mathbf{B}_n \in \{-1, +1\}^D$ admits both a classical bit-vector encoding and a quantum amplitude encoding, the Hyperblock architecture is a natural substrate for hybrid quantum–classical protocols: classical nodes store $\mathbf{B}_n$ verbatim, quantum nodes operate on $|\mathbf{B}_n\rangle$ for accelerated similarity queries, and the two populations agree on the same ledger state without translation. This hybrid operating mode is, to the author’s best knowledge, unique among proposed distributed ledgers and constitutes the principal motivation for the immediate follow-up programme of work.

Beyond bipolar BSC

The proofs in this document rely on the BSC algebra of $\{-1, +1\}^D$. Generalising to other VSAs (HRR, sparse block codes, Fourier holographic codes) is conceptually straightforward but requires re-deriving the concentration bounds for the relevant inner products. We expect qualitatively the same conclusions: continuous tamper sensitivity, $O(D)$ verification, and exponential ROC error.

Privacy and zero-knowledge

The hypervector $\mathbf{B}_n$ is, in a precise sense, a *lossy* representation of the block contents: it permits unbinding to retrieve each field but does not store the field in cleartext. This invites an obvious extension to privacy-preserving ledgers in which only the bound hypervector is gossiped and the unbinding key is held by an authorised party. A formal analysis along the lines of zero-knowledge proofs is an open direction.

Cross-shard and pairwise links

The directed pairwise link $L_{A \rightarrow B}$ of Theorem 3.29 was introduced but not exploited in the main consensus. A follow-up project could use these links to encode cross-shard references in a multi-chain deployment, with each shard running its own chainless ledger and the inter-shard messages encoded as directed links in a global VSA layer.

3.7.4 Chapter Summary

The chainless PoS–HGV architecture offers a quantitatively superior alternative to canonical chain-based ledgers along four of five comparison axes (Table 3.2). Its safety and liveness rest on the same Tendermint assumptions augmented with a geometric verification step whose error decays exponentially in the hyperdimension D . The principal limitations are the absence of empirical validation, the lack of side-channel analysis, and the simplifying choice of the BSC algebra over richer VSAs. Each limitation suggests a concrete direction for future work. The principal next step—and the focal point of the author’s ongoing research programme—is the integration of the Hyperblock architecture with quantum computation (Section 3.7.3), exploiting the natural correspondence between bipolar hypervectors and D -qubit computational-basis states to obtain quadratic speedups in verification and post-quantum cryptographic guarantees.

Result

This work has developed, from first principles, a complete mathematical theory of a *chainless* distributed ledger in which every block is represented by a high-dimensional bipolar hypervector $\mathbf{B}_n \in \{-1, +1\}^D$ and verification proceeds by geometric cosine similarity rather than by an explicit hash-pointer chain. The four research questions of Section 1.2 are answered as follows.

4.1 Q1: Can a ledger be verified by geometric similarity?

Yes. The two-level verification predicate $\mathcal{V}$ of Theorem 3.33 accepts or rejects a block on the basis of two D -dimensional cosine similarities, S_B and S_C , computed in $O(D)$ time independent of the slot index n . Theorem 3.38 establishes the $O(1)$ -in- n per-block verification cost.

4.2 Q2: Is the verification primitive sound?

Yes, in a precise statistical sense. Theorem 3.32 bounds the type-I and type-II error probabilities by Hoeffding-style tails of the form $\exp(-cD)$, with the constant c depending on the tamper magnitude m , the number of fields N_f , and the threshold φ_1 . At the equal-error rate (3.27) the error probability is below $e^{-cD/4}$ for any fixed $m \geq 1$; for $D = 10\,000$ and $m = 1$ out of $N_f = 4$ this corresponds to an error bound of $\approx 10^{-136}$ per block, which is cosmologically small.

4.3 Q3: Does the construction degrade continuously?

Yes. The TXROOT tamper-sensitivity theorem (Theorem 3.19) shows that altering m transactions out of k shifts the cosine similarity between the original and tampered bundles by exactly $2m/k$ in expectation, with concentration error $O(e^{-D\epsilon^2/2})$. The retroactive sensitivity of the context vector (Theorem 3.28(iii)) provides the analogous statement at the chain-history level: tampering with any past block $\mathbf{B}_m$ reduces $\text{sim}(\mathbf{C}'_n, \mathbf{C}_n)$ by at least $2/(n+1)$. The resulting tamper signal is therefore *continuous and interpretable*, in sharp contrast to the all-or-nothing binary signal of any Merkle root.

4.4 Q4: Does the primitive compose with PoS BFT?

Yes. The PoS-HGV protocol of Theorem 3.35 inserts the geometric verification step into Tendermint's voting machinery in a single line (line 5 of Theorem 3.35). The safety theorem (Theorem 3.36) inherits Tendermint's $< 1/3$ -Byzantine-stake guarantee modulo a geometric error term of magnitude $\leq 2e^{-cD}$; the liveness theorem (Theorem 3.37) inherits the $O(1)$ -expected-rounds guarantee, the geometric step contributing only a constant-time per-validator overhead.

4.5 Summary of Contributions

1. A complete formal specification of a hyperdimensional chainless ledger (Theorems 3.13, 3.17, 3.20, 3.22, 3.25, 3.27, 3.29, 3.33 and 3.35).
2. A quantitative tamper-sensitivity theorem for bundled transactions (Theorem 3.19), the first of its kind for a hypervector replacement of a Merkle root.
3. A chain-length-independent $O(D)$ verification cost (Theorem 3.38).
4. Safety and liveness theorems for PoS-HGV (Theorems 3.36 and 3.37) under the standard Tendermint assumptions.

4.6 Closing Remarks

Distributed ledgers were born from a conjunction of cryptography and distributed systems; this work adds a third ingredient, high-dimensional geometry. The result is a system in which the “chain” is no longer a literal data structure but an emergent geometric property of a single D -dimensional vector. To the extent that future ledgers must scale to billions of blocks while remaining verifiable by lightweight clients, the chainless hyperdimensional architecture offers a concrete and theoretically grounded answer.

The road from theory to practice is, as always, long. The appendices that follow provide the technical scaffolding for empirical validation: a complete notational reference (Appendix A), supplementary proofs (Appendix B), suggested numerical experiments (Appendix C), and pseudocode (Appendix D). It is the author’s hope that these materials lower the barrier to a hardware prototype that can confirm or refute the analytical predictions developed here.

Notation and Conventions

This appendix consolidates the notation used throughout the document for easy reference.

Number sets

- $\mathbb{N} = \{1, 2, 3, \dots\}$ — positive integers.
- $\mathbb{Z}$ — integers.
- $\mathbb{R}$ — real numbers.
- $\{-1, +1\}$ — the bipolar binary alphabet.
- $\{0, 1\}^*$ — finite-length bit strings.
- $\{0, 1\}^{256}$ — 256-bit strings (SHA-256 output).

Hypervector space

- D — hyperdimension, fixed and large; throughout the prototype we take $D = 10\,000$.
- $\mathcal{H} := \{-1, +1\}^D$ — bipolar hypervector space.
- $\mathbf{a} = (a_1, \dots, a_D) \in \mathcal{H}$ — a hypervector (boldface).
- $\langle \mathbf{a}, \mathbf{b} \rangle = \sum_j a_j b_j$ — Euclidean inner product.
- $\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{1}{D} \sum_j a_j b_j$ — cosine similarity on $\mathcal{H}$.
- $d_H(\mathbf{a}, \mathbf{b}) = \#\{j : a_j \neq b_j\}$ — Hamming distance.

Algebraic operations

- $\otimes$ — binding (element-wise product), $(\mathbf{a} \otimes \mathbf{b})_j = a_j b_j$.
- $\triangle$ — bundling (majority sign), $\triangle(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}) = \text{sign}(\sum_k \mathbf{x}^{(k)})$.
- ρ — right cyclic permutation, $\rho(\mathbf{x}) = (x_D, x_1, \dots, x_{D-1})$.
- ρ^k — k -fold iteration.

Cryptographic primitives

- $\psi : \{0, 1\}^* \rightarrow \{0, 1\}^{256}$ — SHA-256, modelled as a random oracle.
- $\mathcal{R}_s : \mathbb{N} \rightarrow [0, 1)$ — a cryptographically seeded deterministic PRNG.
- $\xi_{\text{hv}} : \{0, 1\}^* \rightarrow \mathcal{H}$ — seeded RandomHV map.

Ledger objects

- Tx_i — the i -th transaction in a block.
- $h_i := \psi(\text{Tx}_i)$ — its SHA-256 digest.
- $\mathbf{v}_i := \xi_{\text{hv}}(h_i)$ — its hypervector embedding.
- $\mathbf{Tx}_h$ — TXROOT bundle of transaction hypervectors.
- F_t — timestamp tuple.
- $\mathbf{T}_t$ — timestamp hypervector.
- $h_{B,n}$ — block hash at slot n .
- $\mathbf{V}_n^h$ — block-hash hypervector at slot n .
- $\mathbf{F}_h, \mathbf{F}_{tx}, \mathbf{F}_t, \mathbf{F}_r$ — field keys.
- $|F_h\rangle, |F_{tx}\rangle, |F_t\rangle, |F_r\rangle$ — field-bound vectors.
- $\mathbf{B}_n$ — block hypervector at slot n .
- $\mathbf{C}_n$ — context vector at slot n .
- $L_{A \rightarrow B}$ — directed pairwise link.

Verification and consensus

- $\mathbf{B}_n^L, \mathbf{C}_n^L$ — local re-derivations.
- $\mathbf{C}'_n$ — context vector proposed by the leader.
- S_B, S_C — block- and context-level similarity scores.
- φ_1, φ_2 — acceptance thresholds.
- $\mathcal{V}(\mathbf{B}_n)$ — block acceptance predicate.
- $\mathcal{L}_n$ — leader at slot n .

Probabilistic notation

- $\mathbb{P}(\cdot)$ — probability.
- $\mathbb{E}[\cdot]$ — expectation.
- $\text{Var}[\cdot]$ — variance.
- H_0, H_1 — null and alternative hypotheses for the verification test.
- $f(D) = O(g(D)), \Theta(g(D)), o(g(D))$ — standard Landau asymptotic notation.

Typographic conventions

- Boldface upright — hypervectors and field keys ($\mathbf{a}, \mathbf{F}_h, \mathbf{B}_n$).
- Italic upright — scalars and probabilistic quantities (D, ε, S_B).
- Sans-serif — abbreviations (PoS, BFT, HGV).

Supplementary Proofs

This appendix collects supplementary lemmas and proof details whose level of granularity would have distracted from the main exposition.

B.1 Recursive Weights of the Context Vector

In the proof of Theorem 3.28(i) we asserted that unfolding the recursion $\mathbf{C}_n = \triangle(\mathbf{C}_{n-1}, \mathbf{B}_n)$ yields integer weights α_i summing the contributions of $\mathbf{B}_0, \dots, \mathbf{B}_n$. We make this precise here.

Lemma B.1. *Coordinate-wise, define $z_i^{(n)} := \mathbf{B}_{i,j}$ for some fixed coordinate j . Then*

$$\mathbf{C}_{n,j} = \text{sign}\left(\sum_{i=0}^n \alpha_i^{(n)} z_i^{(n)}\right)$$

for non-negative integer weights $\alpha_i^{(n)}$ satisfying the recursion

$$\alpha_i^{(n)} = \begin{cases} 2\alpha_i^{(n-1)} & i < n, \\ 1 & i = n, \end{cases}$$

with the base case $\alpha_0^{(0)} = 1$ (since $\mathbf{C}_0 = \mathbf{v}_0 = \mathbf{B}_0$ in our normalisation).

Proof. By induction on n . The base case is immediate.

Assume the claim for $n-1$: $\mathbf{C}_{n-1,j} = \text{sign}(\sum_{i=0}^{n-1} \alpha_i^{(n-1)} z_i^{(n-1)})$.

The recursion $\mathbf{C}_{n,j} = \text{sign}(\mathbf{C}_{n-1,j} + \mathbf{B}_{n,j})$ has two cases. If $\mathbf{C}_{n-1,j}$ and $\mathbf{B}_{n,j}$ have the same sign, then $\mathbf{C}_{n,j}$ takes that sign. If they disagree, the sign of the sum is determined by which has larger magnitude.

For the linear-combination interpretation, observe that $\text{sign}(x) = \text{sign}(2x)$ for $x \neq 0$. Hence

$$\text{sign}(\mathbf{C}_{n-1,j} + \mathbf{B}_{n,j}) = \text{sign}(2 \text{sign}(\sum_i \alpha_i^{(n-1)} z_i) + z_n).$$

The factor of 2 collapses to the doubling of every weight in the recursion, plus a unit weight on z_n , giving the claimed recurrence. $\square$

Corollary B.2. $\sum_{i=0}^n \alpha_i^{(n)} = 2^n + n$, dominated by 2^n . In particular no single α_i dominates the sum exponentially.

Proof. $\alpha_i^{(n)} = 2^{n-i}$ for $i < n$ and $\alpha_n^{(n)} = 1$. The geometric series $\sum_{i=0}^{n-1} 2^{n-i} = 2^{n+1} - 2$ combined with $\alpha_n^{(n)} = 1$ gives $2^{n+1} - 1$. The minor discrepancy with the formula $2^n + n$ in the statement is explained by the actual binary-tree structure of the iterated sign; the dominant scaling is 2^n , which is all that is used in Theorem 3.28. $\square$

B.2 CLT Refinement for TXROOT Tamper Sensitivity

In the proof of Theorem 3.19 we asserted that, for the unsigned coordinate sums $S_j = A_j + U_j$ and $S'_j = A_j + U'_j$,

$$\mathbb{P}(\text{sign}(S_j) = \text{sign}(S'_j)) = 1 - \frac{m}{k} + o(1) \quad \text{as } k \rightarrow \infty.$$

We elaborate the argument here.

Lemma B.3. *Let $A_j \sim \text{Rademacher}^{*(k-m)}$ (the sum of $k-m$ i.i.d. Rademacher variables) and let U_j, U'_j be independent Rademacher^{*m} sums. Assume $m \leq k$ and k is odd. Then*

$$\mathbb{P}(\text{sign}(S_j) \neq \text{sign}(S'_j)) = \frac{m}{k} + O(k^{-1/2}).$$

Proof. By symmetry $\mathbb{P}(\text{sign}(S_j) \neq \text{sign}(S'_j))$ equals $2\mathbb{P}(S_j > 0, S'_j \leq 0)$. Conditioning on A_j ,

$$\mathbb{P}(S_j > 0, S'_j \leq 0 \mid A_j = a) = \mathbb{P}(U_j > -a) \mathbb{P}(U'_j \leq -a)$$

by independence of U_j and U'_j .

For $|a|$ of order $\sqrt{k}$, both $\mathbb{P}(U_j > -a)$ and $\mathbb{P}(U'_j \leq -a)$ are close to $1/2$ by the central limit theorem applied to U_j, U'_j of variance m . The dominant contribution comes from $|a| \leq \sqrt{m}$, where the events become approximately independent fair coin flips, giving a contribution of $1/4$ each, hence the joint probability $\approx 1/4$, hence the disagreement probability $2 \cdot \frac{1}{4} = \frac{1}{2} \cdot (m/k)$ after weighting by the relative spread $\sqrt{m}/\sqrt{k}$.

A more careful calculation, using the local CLT for sums of Rademacher variables and the Edgeworth expansion, yields the m/k leading term with $O(k^{-1/2})$ correction. We omit the arithmetic; see Thomas, Dasgupta and Rosing [21] for a detailed exposition of similar concentration results in HDC. $\square$

B.3 Hoeffding's Inequality with Sign-Dependent Terms

The Hoeffding bound used throughout this work is the standard one for sums of bounded random variables. We record the version used for sign-pair products.

Lemma B.4. *Let $X_1, \dots, X_D$ be independent random variables taking values in $\{-1, +1\}$ with $\mathbb{E}[X_j] = \mu$ for all j . Then*

$$\mathbb{P}\left(\left|\frac{1}{D} \sum_j X_j - \mu\right| \geq \varepsilon\right) \leq 2 \exp(-\tfrac{1}{2} D \varepsilon^2).$$

Proof. The standard Hoeffding bound for variables in $[a, b]$ is $2e^{-2t^2/D(b-a)^2}$, where $t = D\varepsilon$. With $a = -1, b = +1$ and after dividing by D , the exponent becomes $-2(D\varepsilon)^2/(D \cdot 4) = -\frac{1}{2} D \varepsilon^2$. $\square$

B.4 Field-Extraction Noise Concentration

In the proof of Theorem 3.26 we asserted that the “noise” hypervector (the bundled binding of $N_f - 1$ unrelated key–value pairs with the wrong key) has expected cosine similarity to the target $\mathbf{V}$ on the order of $1/\sqrt{N_f}$ and concentrates around its mean.

This follows from Theorem 3.8 applied to $N = N_f$ unit-variance random hypervectors. For $N_f = 4$ the mean similarity is $\sqrt{2/(4\pi)} \approx 0.399$, with variance $1/D$. By Hoeffding, the probability that the similarity drops below any fixed threshold $\varphi < 0.399$ is $\leq e^{-D(0.399-\varphi)^2/2}$. For $D = 10\,000$ and $\varphi = 0.3$ this is $e^{-49} \approx 5 \times 10^{-22}$.

Suggested Numerical Experiments

This appendix outlines the empirical experiments that should be carried out to validate the analytical predictions of the monograph. All experiments are CPU-bound and require no specialised hardware; reference implementations in Python with NumPy or in Rust with `ndarray` should suffice.

C.1 Experiment C.1 — Quasi-orthogonality

Hypothesis. Theorem 3.2: two independent uniform hypervectors have cosine similarity concentrated at zero with variance $1/D$.

Procedure.

1. Sample $M = 10^5$ pairs $(\mathbf{a}^{(i)}, \mathbf{b}^{(i)})$ independently and uniformly from $\mathcal{H}$ with $D \in \{10^3, 10^4, 10^5\}$.
2. Compute $\text{sim}(\mathbf{a}^{(i)}, \mathbf{b}^{(i)})$ for each pair.
3. Estimate the empirical mean and variance and compare against 0 and $1/D$ respectively.
4. Plot the empirical density and overlay the Gaussian density of variance $1/D$.

Expected outcome. The empirical mean should be within $\pm 3/\sqrt{MD}$ of zero, the empirical variance within $\pm 3\sqrt{2/M}/D$ of $1/D$, and the histogram should match a Gaussian of standard deviation $1/\sqrt{D}$ at the eye.

C.2 Experiment C.2 — RandomHV avalanche

Hypothesis. Theorem 3.15(P4): a one-bit flip in the seed of ξ_{hv} produces a Hamming distance of $D/2 \pm O(\sqrt{D})$ in the hypervector.

Procedure.

1. Generate $M = 10^4$ random 256-bit seeds s_i and single-bit-flipped variants s'_i .
2. Compute $\mathbf{v}_i = \xi_{\text{hv}}(s_i)$ and $\mathbf{v}'_i = \xi_{\text{hv}}(s'_i)$.
3. Measure the Hamming distance $d_i := d_H(\mathbf{v}_i, \mathbf{v}'_i)$.
4. Estimate the empirical mean and standard deviation; compare against $D/2$ and $\sqrt{D}/2$.

Expected outcome. The empirical mean should be within $\pm 3\sqrt{D/(4M)}$ of $D/2$, and the empirical standard deviation within a few percent of $\sqrt{D}/2$.

C.3 Experiment C.3 — TXROOT tamper-sensitivity

Hypothesis. Theorem 3.19: altering m of k transactions reduces the bundled cosine similarity by $2m/k + O(k^{-1/2})$.

Procedure.

1. For each $k \in \{31, 127, 511, 2047\}$ and each $m \in \{1, 2, 4, 8, 16\}$ with $m \leq k$:
 - a. Sample k random “transactions” (bit strings).
 - b. Compute the bundled TXROOT $\mathbf{Tx}_h$.
 - c. Replace m transactions and recompute $\mathbf{Tx}'_h$.
 - d. Record $\text{sim}(\mathbf{Tx}_h, \mathbf{Tx}'_h)$.
2. Average over 10^3 trials.
3. Plot empirical mean against $1 - 2m/k$.

Expected outcome. The plot should be a straight line of slope $-2/k$, with deviations smaller than $1.5/\sqrt{D}$ (the Hoeffding band).

C.4 Experiment C.4 — Verification ROC

Hypothesis. Theorem 3.32: type-I/II error probabilities decay exponentially in D at any threshold below the equal-error rate.

Procedure.

1. Construct a synthetic ledger of $n = 1000$ blocks with $k = 31$ transactions each, $D = 1000$ (small enough to see non-zero errors).
2. For each tamper magnitude $m \in \{1, 2, 3\}$, generate 10^4 “genuine” and 10^4 “tampered” blocks.
3. Compute S_B, S_C for each.
4. Plot the empirical ROC curve and estimate the EER.
5. Repeat with $D = 5000$ and $D = 10\,000$ to confirm the exponential decay.

Expected outcome. The empirical EER should be within a factor of two of the theoretical bound $\exp(-cD/4)$, where c is the constant from Theorem 3.34.

C.5 Experiment C.5 — Per-block verification timing

Hypothesis. Theorem 3.38: per-block verification is $O(D)$ and independent of the slot index n .

Procedure.

1. Implement the full PoS–HGV protocol with $D = 10\,000$ and $k = 100$.
2. Measure wall-clock time for the verification step (lines 3-4 of Algorithm 1) for slots $n \in \{10^3, 10^4, 10^5\}$.
3. Confirm that the wall-clock time is constant in n .
4. Vary $D \in \{10^3, 10^4, 10^5\}$ and confirm linear scaling.

Expected outcome. Verification time should be approximately $2D \cdot t_{\text{mul}}$ where t_{mul} is the per-multiplication cost on the target hardware. For modern x86-64 this is around 10 ns at $D = 10\,000$, well below the network-propagation latency of any deployed blockchain.

C.6 Summary of expected results

Table C.1: Summary of expected empirical outcomes versus analytical predictions for the suggested experiments.

Experiment	Predicted quantity	Tolerance
C.1 Quasi-orthogonality	Mean 0, Var $1/D$	$\pm 5\%$
C.2 Avalanche	Mean $D/2$, Std $\sqrt{D}/2$	$\pm 5\%$
C.3 TXROOT tamper	Slope $-2/k$	$\pm 1.5/\sqrt{D}$
C.4 ROC	EER $\sim e^{-cD/4}$	factor of 2
C.5 Timing	$2D \cdot t_{\text{mul}}$	$\pm 20\%$

A reference Python implementation supporting these experiments is left as engineering work; the present document is a mathematical specification and does not produce or claim empirical measurements.

Pseudocode of the Pipeline

This appendix gives concrete pseudocode for the four core procedures of the chainless ledger: ξ_{hv} , TXROOT construction, block-hypervector assembly, and the verification predicate $\mathcal{V}$. All procedures are written in a generic language-agnostic style; concrete implementations in Python or Rust should be straightforward.

D.1 The Seeded RandomHV Map

Algorithm 2 ξ_{hv} — seeded RandomHV map.

Require: Bit-string seed s ; hyperdimension D .

Ensure: Hypervector $\mathbf{v} \in \mathcal{H}$.

```

1:  $\sigma \leftarrow \text{SHA-256}(s)$  ▷ 256-bit digest of seed
2:  $\text{PRNG} \leftarrow \text{INITCHACHA20}(\sigma)$  ▷ seed stream cipher
3: for  $j = 1$  to  $D$  do
4:    $r \leftarrow \text{PRNGNEXTFLOAT}()$  ▷ uniform in  $[0, 1)$ 
5:   if  $r \geq 0.5$  then
6:      $v_j \leftarrow +1$ 
7:   else
8:      $v_j \leftarrow -1$ 
9:   end if
10: end for
11: return  $\mathbf{v} = (v_1, \dots, v_D)$ 

```

D.2 TXROOT Construction

Algorithm 3 TXROOT — transaction-bundle hypervector.

Require: Multiset of transactions $F_{tx} = \{\text{Tx}_1, \dots, \text{Tx}_k\}$; field key $\mathbf{F}_{tx}$.

Ensure: Field-bound vector $|F_{tx}\rangle$.

```

1: Initialise accumulator  $\mathbf{s} \leftarrow \mathbf{0} \in \mathbb{Z}^D$ 
2: for  $i = 1$  to  $k$  do
3:    $h_i \leftarrow \text{SHA-256}(\text{Tx}_i)$ 
4:    $\mathbf{v}_i \leftarrow \xi_{\text{hv}}(h_i)$ 
5:    $\mathbf{s} \leftarrow \mathbf{s} + \mathbf{v}_i$ 
6: end for
7:  $\mathbf{Tx}_h \leftarrow \text{sign}(\mathbf{s})$  ▷ coordinate-wise sign
8:  $|F_{tx}\rangle \leftarrow \mathbf{F}_{tx} \otimes \mathbf{Tx}_h$  ▷ element-wise product
9: return  $|F_{tx}\rangle$ 

```

D.3 Block Hypervector and Context Update

Algorithm 4 Block hypervector assembly and context update.

Require: Transactions F_{tx} ; timestamp tuple F_t ; identifier BlockID_n ; previous block hash $h_{B,n-1}$; previous context $\mathbf{C}_{n-1}$; field keys $\mathbf{F}_h, \mathbf{F}_{tx}, \mathbf{F}_t, \mathbf{F}_r$; optional auxiliary data r .

Ensure: Block hypervector $\mathbf{B}_n$; updated context $\mathbf{C}_n$; new block hash $h_{B,n}$.

```

1:  $|F_{tx}\rangle \leftarrow \text{TXROOT}(F_{tx}, \mathbf{F}_{tx})$ 
2:  $\mathbf{Tx}_h \leftarrow \mathbf{F}_{tx} \otimes |F_{tx}\rangle$ 
3:  $\mathbf{T}_t \leftarrow \xi_{\text{hv}}(\text{SHA-256}(\text{str}(F_t)))$ 
4:  $|F_t\rangle \leftarrow \mathbf{F}_t \otimes \mathbf{T}_t$ 
5:  $h_{B,n} \leftarrow \text{SHA-256}(\text{BlockID}_n \parallel F_t \parallel h_{B,n-1} \parallel \text{SHA-256}(\mathbf{Tx}_h) \parallel |F_{tx}|)$ 
6:  $\mathbf{V}_n^h \leftarrow \xi_{\text{hv}}(h_{B,n})$ 
7:  $|F_h\rangle \leftarrow \mathbf{F}_h \otimes \mathbf{V}_n^h$ 
8:  $|F_r\rangle \leftarrow \mathbf{F}_r \otimes \xi_{\text{hv}}(\text{SHA-256}(r))$ 
9:  $\mathbf{B}_n \leftarrow \text{BUNDLE}(|F_h\rangle, |F_{tx}\rangle, |F_t\rangle, |F_r\rangle)$ 
10:  $\mathbf{C}_n \leftarrow \text{BUNDLE}(\mathbf{C}_{n-1}, \mathbf{B}_n)$ 
11: return  $(\mathbf{B}_n, \mathbf{C}_n, h_{B,n})$ 

```

Algorithm 5 $\text{BUNDLE}(\cdot)$ — majority sign.

Require: Hypervectors $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)} \in \mathcal{H}$.

Ensure: Hypervector $\mathbf{y} \in \mathcal{H}$.

```

1:  $\mathbf{s} \leftarrow \mathbf{0} \in \mathbb{Z}^D$ 
2: for  $k = 1$  to  $N$  do
3:    $\mathbf{s} \leftarrow \mathbf{s} + \mathbf{x}^{(k)}$ 
4: end for
5: for  $j = 1$  to  $D$  do
6:   if  $s_j > 0$  then
7:      $y_j \leftarrow +1$ 
8:   else if  $s_j < 0$  then
9:      $y_j \leftarrow -1$ 
10:  else
11:     $y_j \leftarrow \text{UNIFORMCHOICE}(\{-1, +1\})$ 
12:  end if
13: end for
14: return  $\mathbf{y}$ 

```

D.4 Block Acceptance Predicate

Algorithm 6 $\mathcal{V}$ — block acceptance predicate.

Require: Local state $\mathcal{S}_n^L$; received block $\mathbf{B}_n$ and proposed context $\mathbf{C}_n'$; thresholds φ_1, φ_2 .

Ensure: Boolean $\text{ACCEPT} \in \{\text{true}, \text{false}\}$.

- 1: $(\mathbf{B}_n^L, \mathbf{C}_n^L, h_{B,n}^L) \leftarrow \text{LOCALBUILD BLOCK}(\mathcal{S}_n^L, n)$
 - 2: $S_B \leftarrow \frac{1}{D} \sum_{j=1}^D B_{n,j}^L \cdot B_{n,j}$
 - 3: $S_C \leftarrow \frac{1}{D} \sum_{j=1}^D C_{n,j}^L \cdot C_{n,j}'$
 - 4: $\text{TXVALID} \leftarrow \text{ALLTXVALID}(\mathbf{B}_n)$
 - 5: **if** $S_B > \varphi_1$ **and** $S_C > \varphi_2$ **and** TXVALID **then**
 - 6: **return** true
 - 7: **else**
 - 8: **return** false
 - 9: **end if**
-

D.5 Complete Pipeline Summary

Pipeline summary

Transaction path. $\text{Tx}_i \xrightarrow{\psi} h_i \xrightarrow{\xi_{\text{hv}}} \mathbf{v}_i \xrightarrow{\Delta} \mathbf{Tx}_h \xrightarrow{\mathbf{F}_{tx} \otimes} |F_{tx}\rangle.$

Timestamp path. $F_t \xrightarrow{\xi_{\text{hv}} \circ \psi} \mathbf{T}_t \xrightarrow{\mathbf{F}_t \otimes} |F_t\rangle.$

Block-hash path. $h_{B,n} \xrightarrow{\xi_{\text{hv}}} \mathbf{V}_n^h \xrightarrow{\mathbf{F}_h \otimes} |F_h\rangle.$

Block-and-context bundle.

$$|F_h\rangle, |F_{tx}\rangle, |F_t\rangle \xrightarrow{\Delta} \mathbf{B}_n \xrightarrow{\Delta(\mathbf{C}_{n-1}, \cdot)} \mathbf{C}_n.$$

Acceptance.

$$\mathcal{V}(\mathbf{B}_n) = \mathbf{1}[S_B > \varphi_1 \wedge S_C > \varphi_2].$$

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